

BARCODE LABEL.65X65
BC2
BARCODE LABEL.65X65

A	DV Change of topology for the backlight power supply	CN0008076	11-29-2023	FY	CZ	
A	ED initial Release	CN0001912	07-30-2023	FY	CZ	
Rev.	type of change	change-no.	date of change	aith.	checked	

 Copyright reserved	Wiring modifications reserved	name		
		date		
		created	11-29-2023	
		checked	11-29-2023	
schematic SCHEM 4010 MAIN C143	FCA FRONT PASSENGER DISPLAY MODULE	Sheet	Sheets of	Sample
		1	31	DV1
		Type	Variant	Part Number
		C143	C143	4106088
substitute for:		substitute by:		

BT

PWR

PWR

UG3V3

L5700

600ohm@100MHz
0.165ohm DCR @ 500mA

220

202

100n

10p

C C5703

C C5706

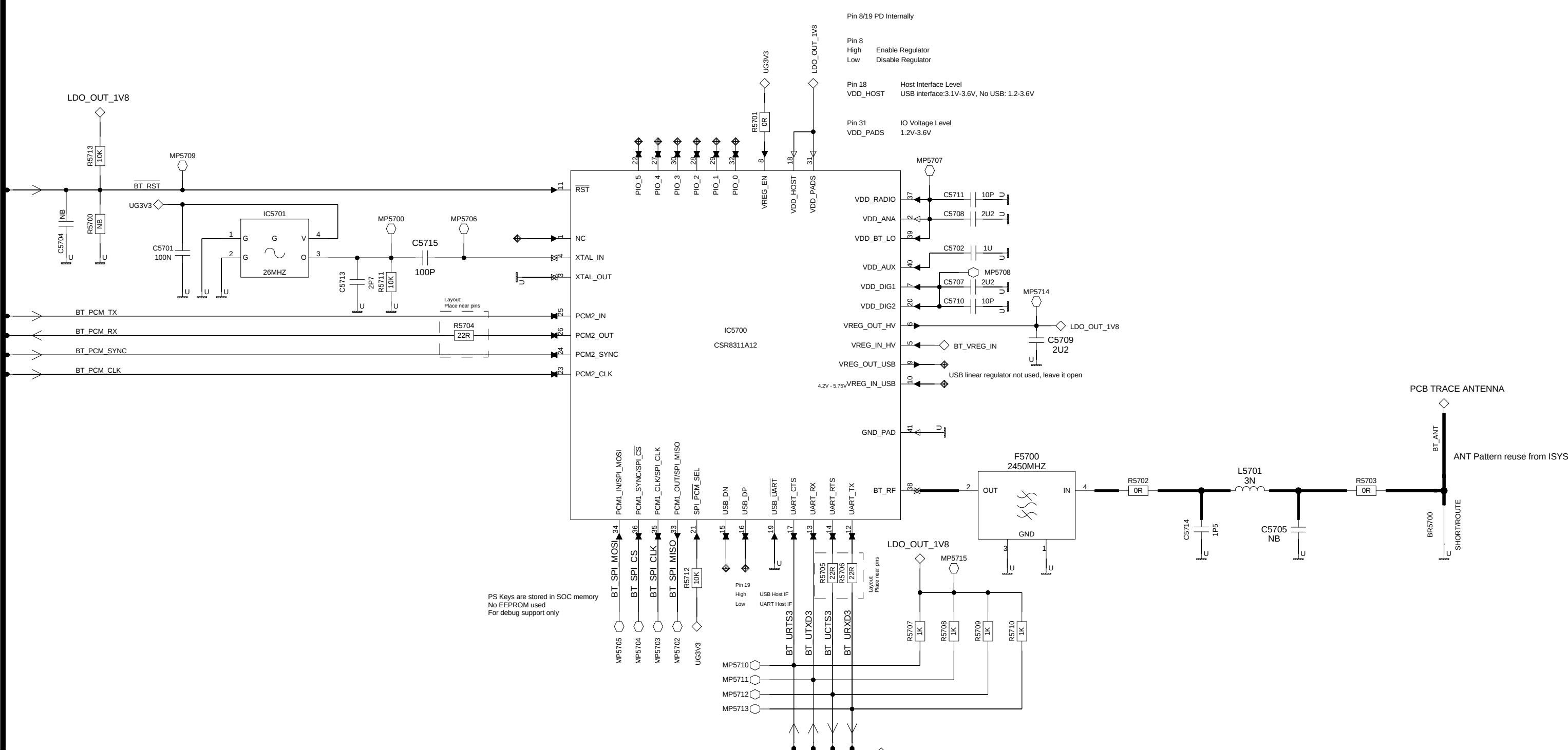
C C5700

C C5712

MP5701

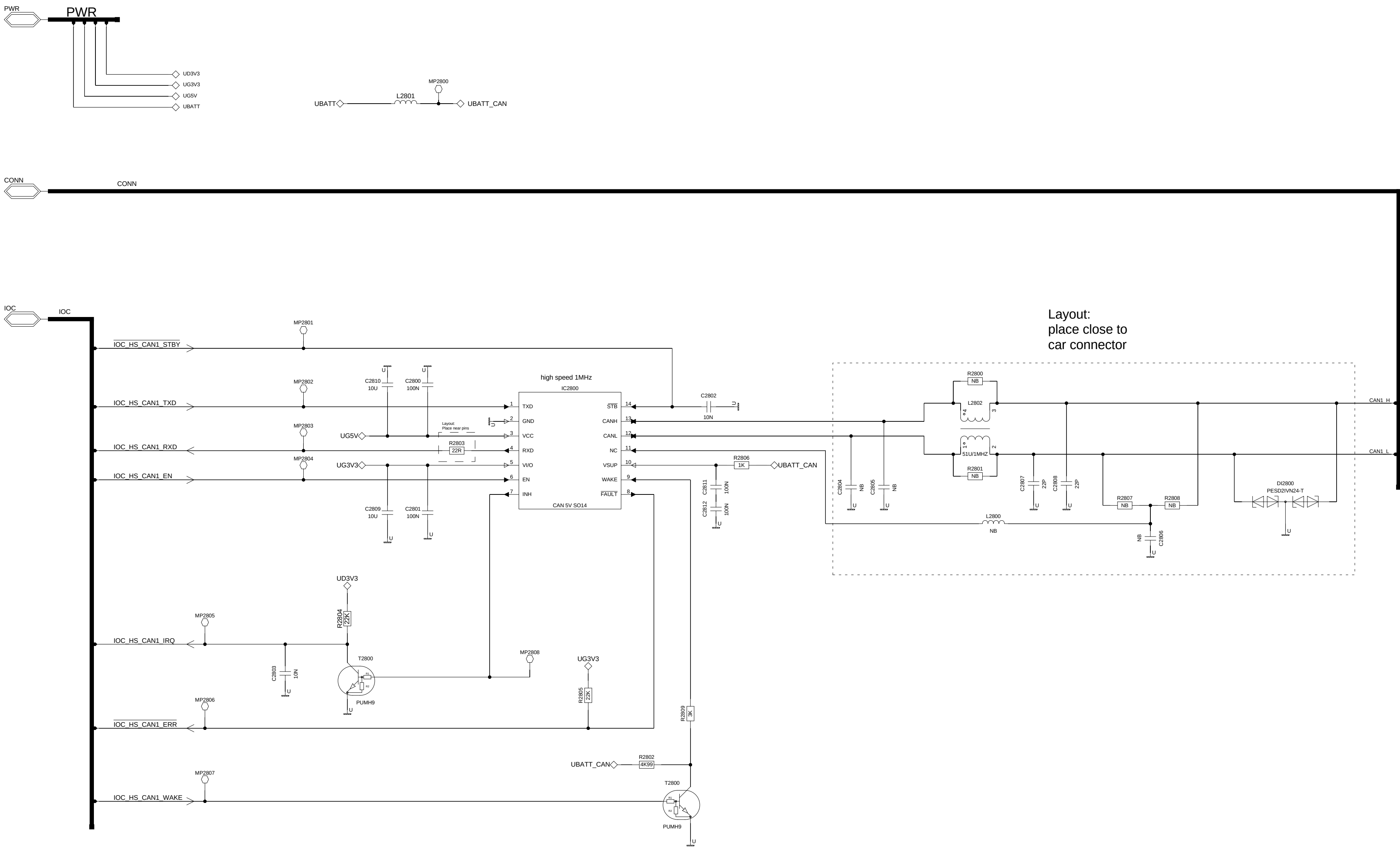
BT_VREG_I

100mA Max.



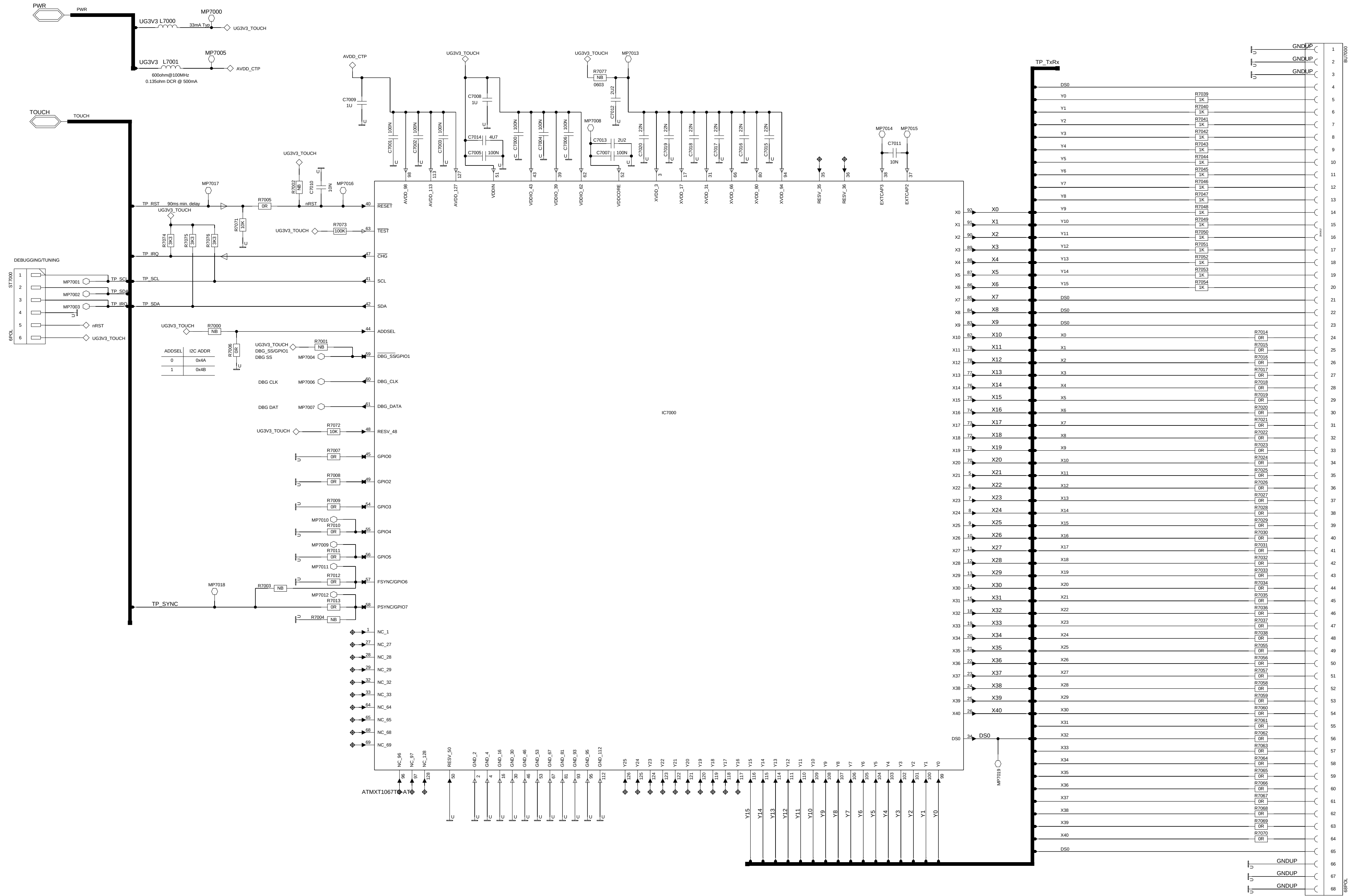
Rev.	type of change		change-no.	date of change	auth. checked
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				created	
				checked	
schematic			Sheet 2	Sheets of 31	Sample
4010_Bluetooth			Type	Variant	Part Number
substitute for:		substitute by:		C143	

CAN (2800-2899)

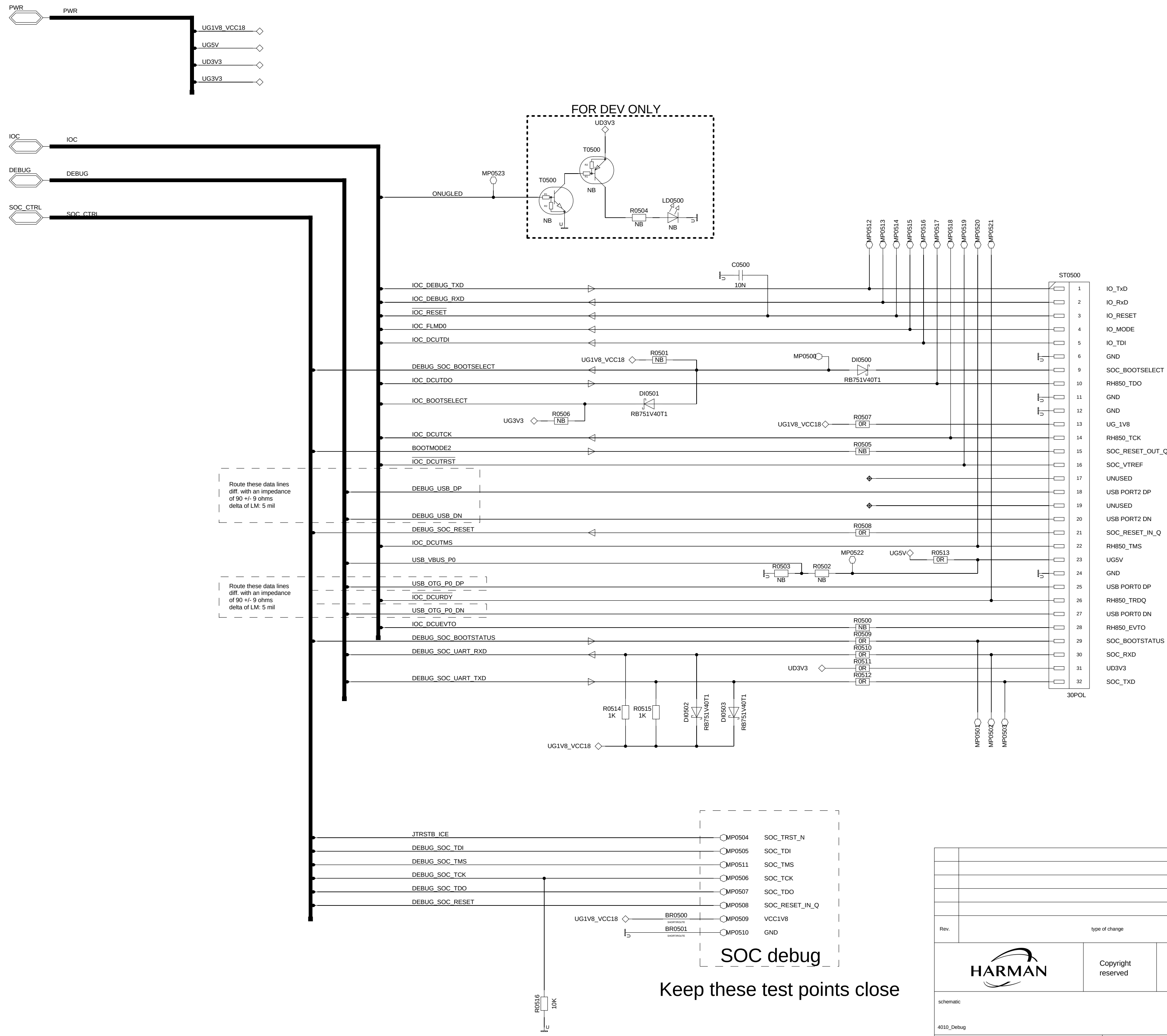


Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 3	Sheets of 31	Sample
4010_CAN			Type	Variant	Part Number
substitute for:		substitute by:			

CTP (7000-7099)



Rev.	type of change		change-no.	date of change	auth.	checked
 HARMAN			date		name	
			created			
			checked			
schematic:			Sheet 4	Sheets of 31	Sample	
4100_CTP			Type	Variant C143	Part Number	
substitute for:			substitute by:			

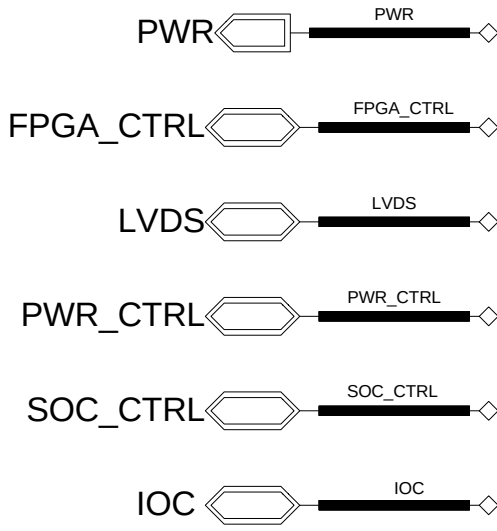


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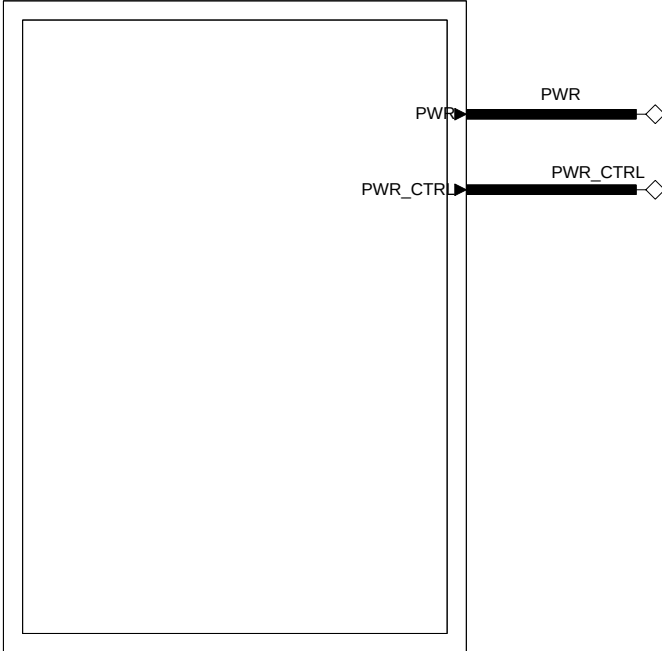
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				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_Debug			5	31	
substitute for:		substitute by:	Type	Variant	Part Number

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_ETH_PHY			6	31	
			Type	Variant	Part Number
substitute for:		substitute by:			



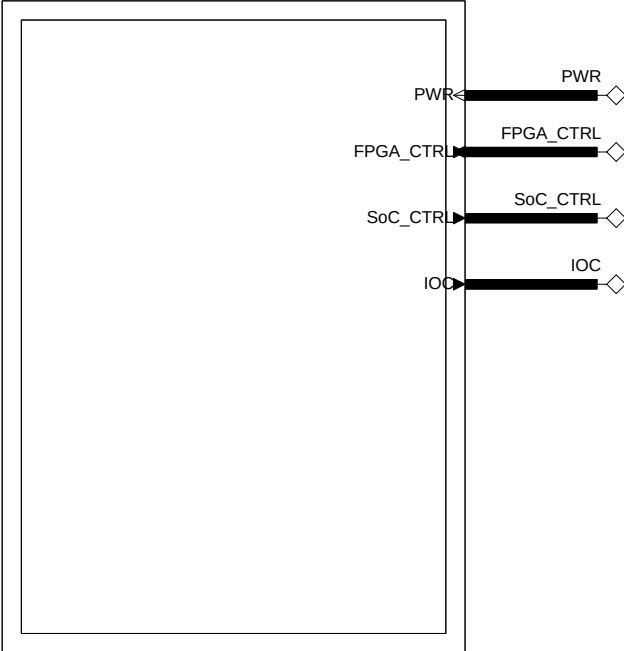
FPGA_POWER



PWR

PWR_CTRL

FPGA_CONF



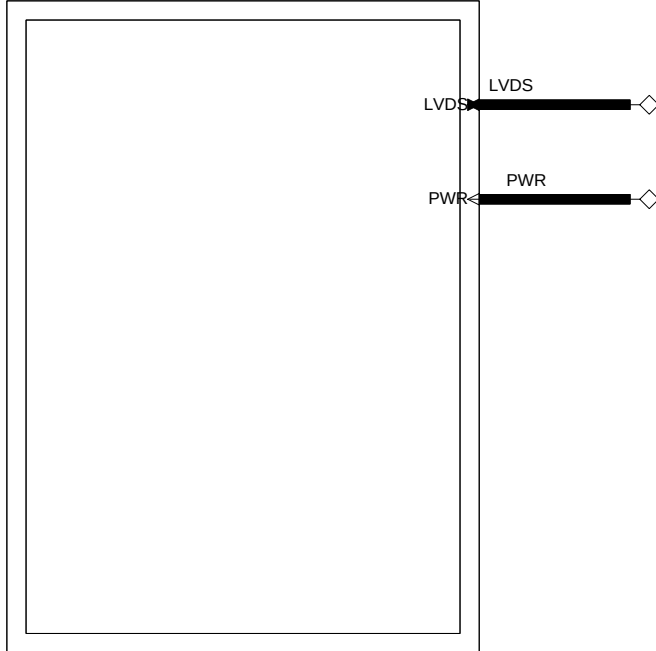
PWR

FPGA_CTRL

SOC_CTRL

IOC

FPGA_VIDEO

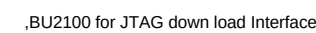
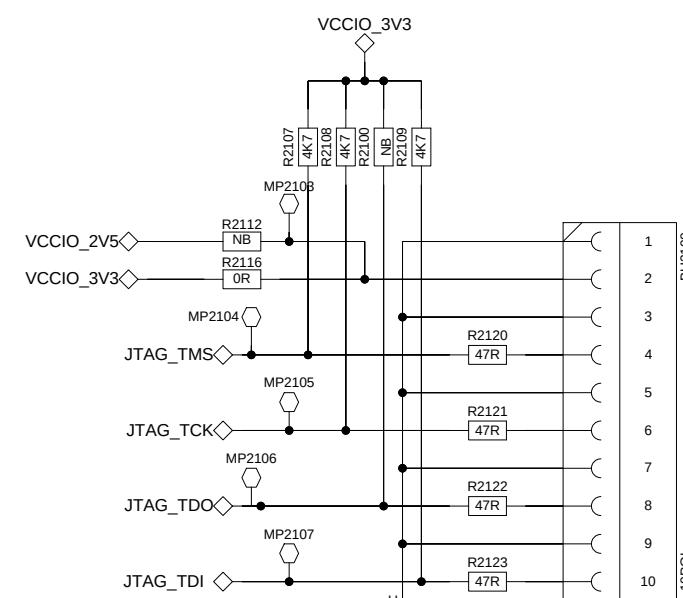
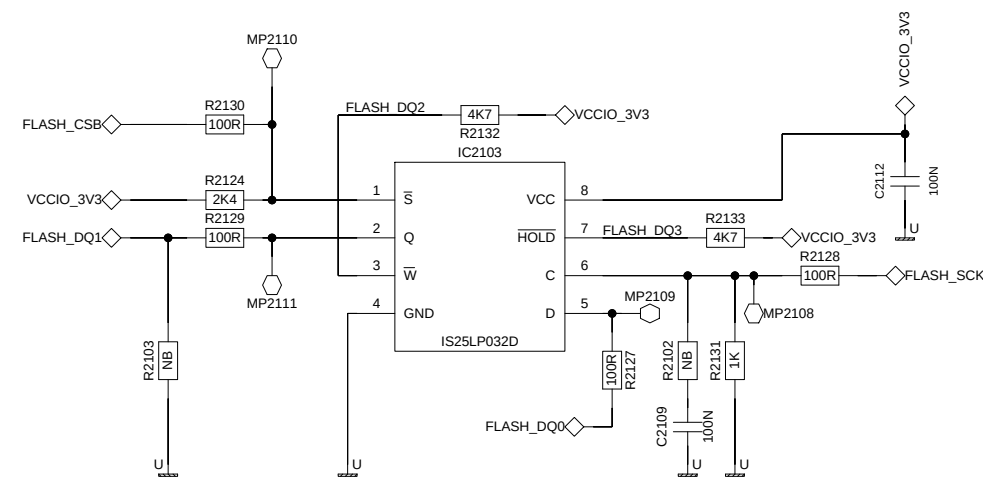
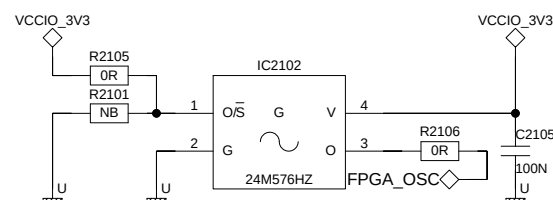
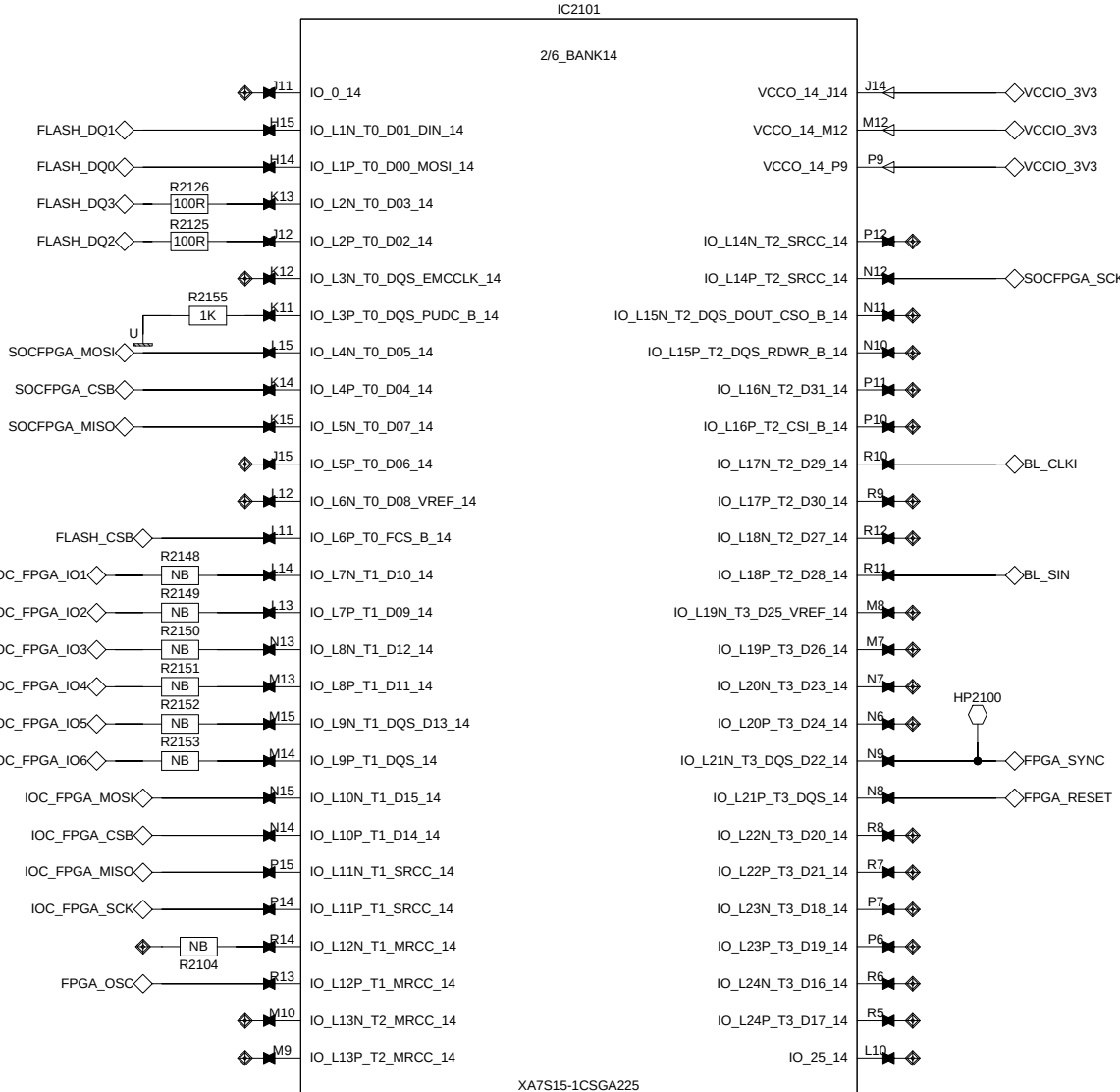
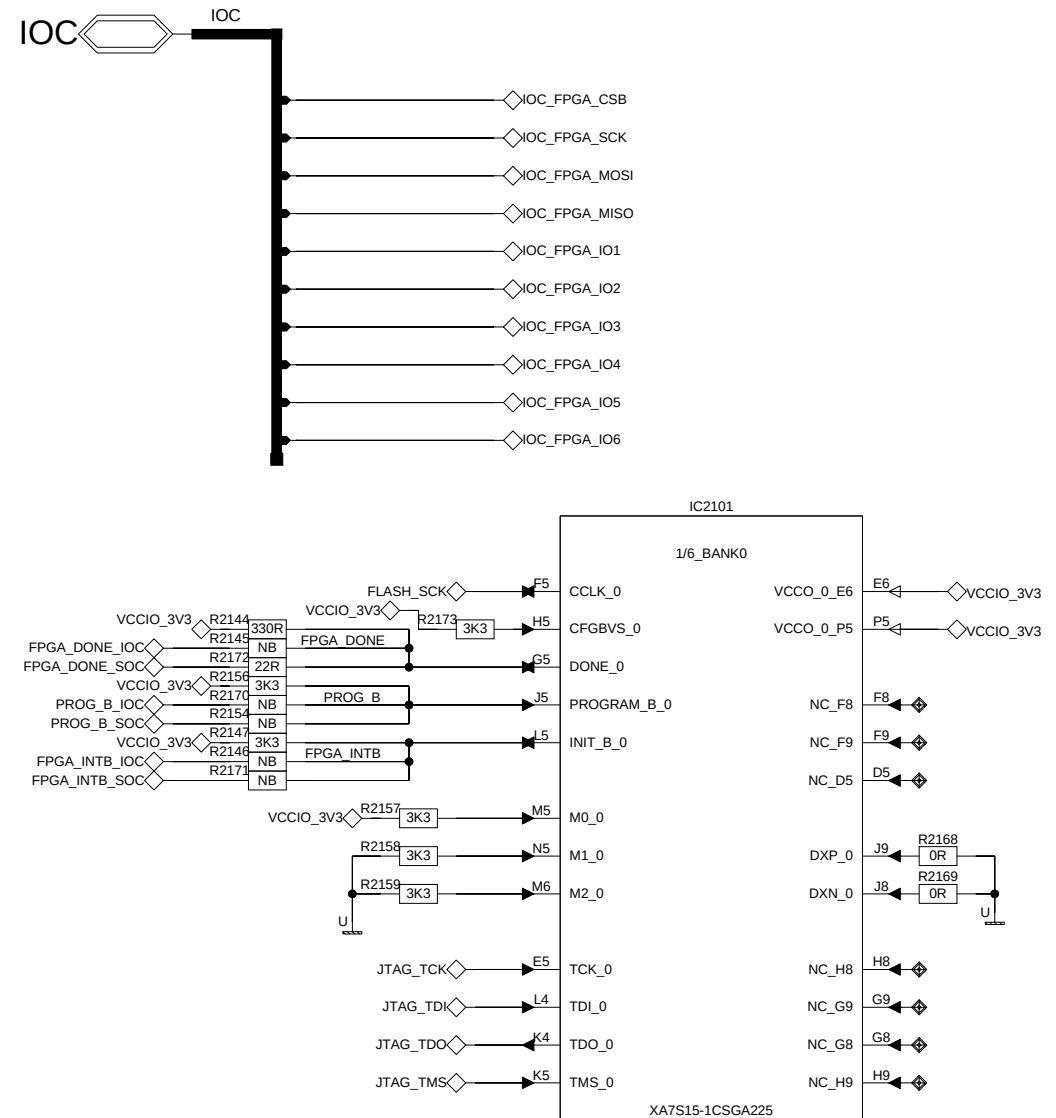
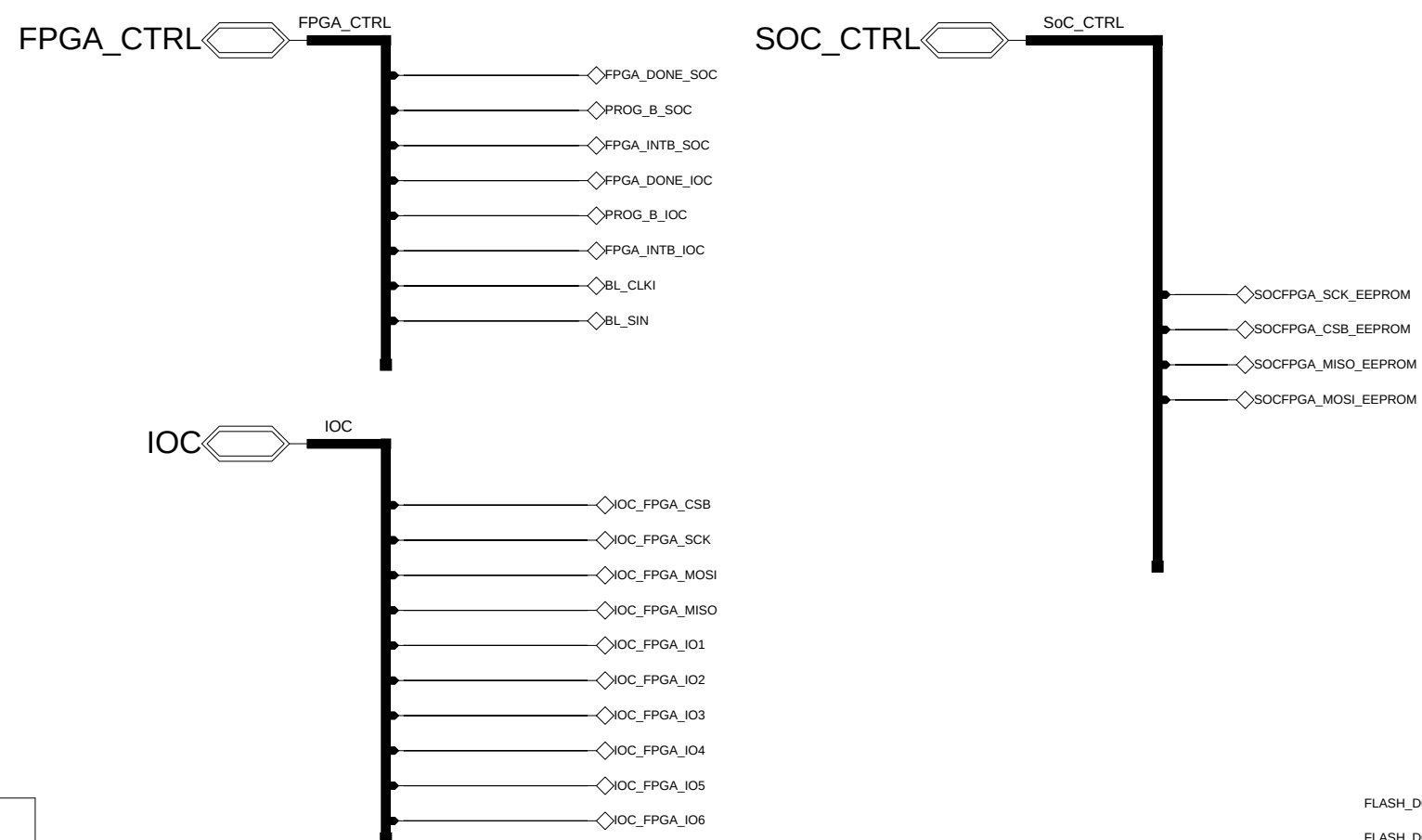


LVDS

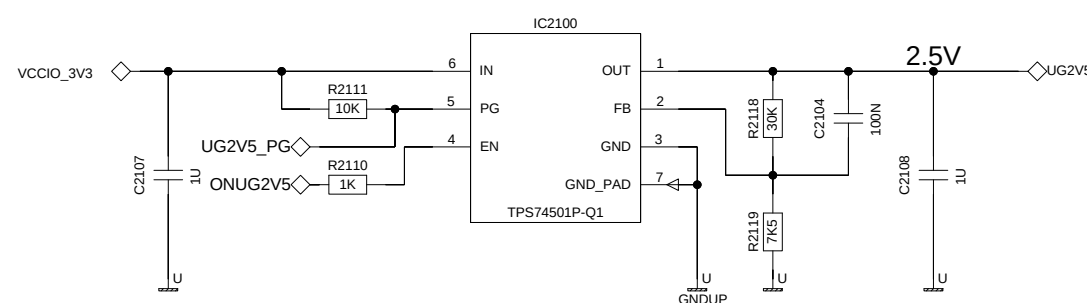
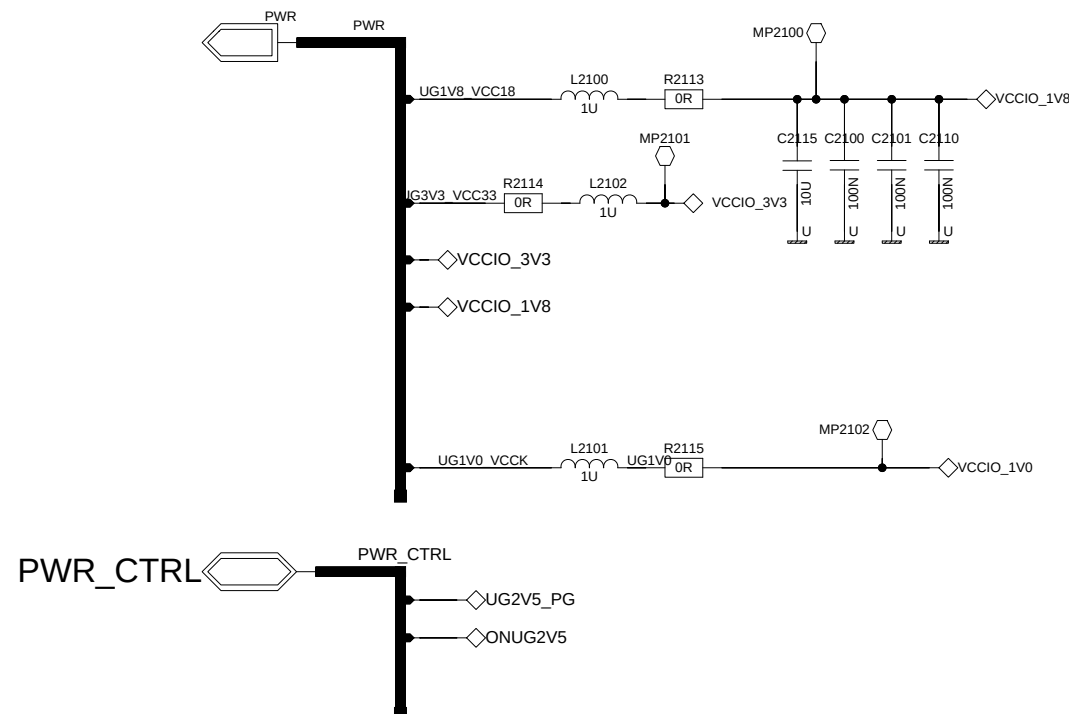
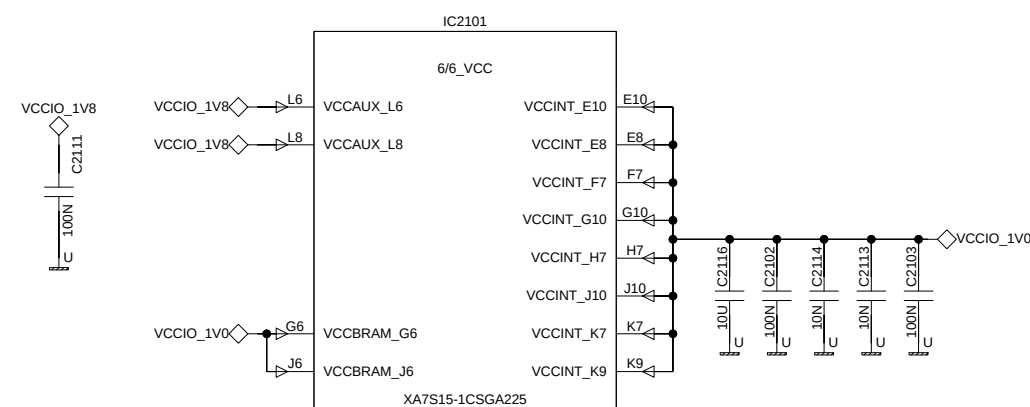
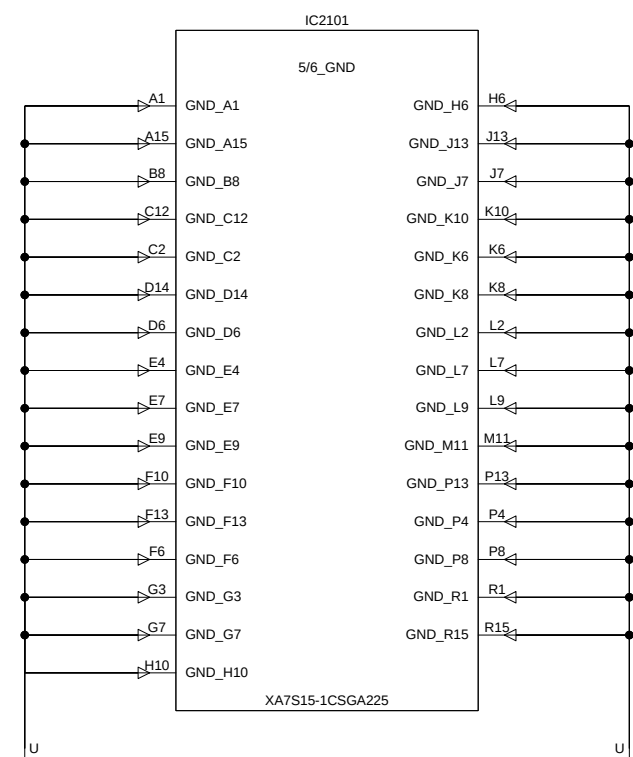
PWR

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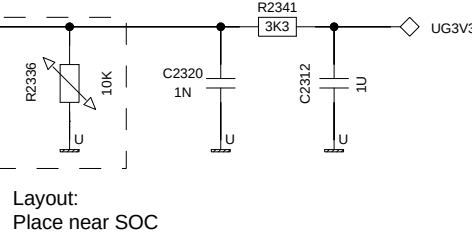
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	Copyright reserved	Writing modifications reserved	date		name		
			created				
			checked				
schematic							
4030_FPGA							
substitute for:		substitute by:					
		Sheet 7		Sheets of 31		Sample	
		Type		Variant C143		Part Number	



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 8	Sheets of 31	Sample
FPGA_CONF			Type	Variant	Part Number
substitute for:		substitute by:		C143	



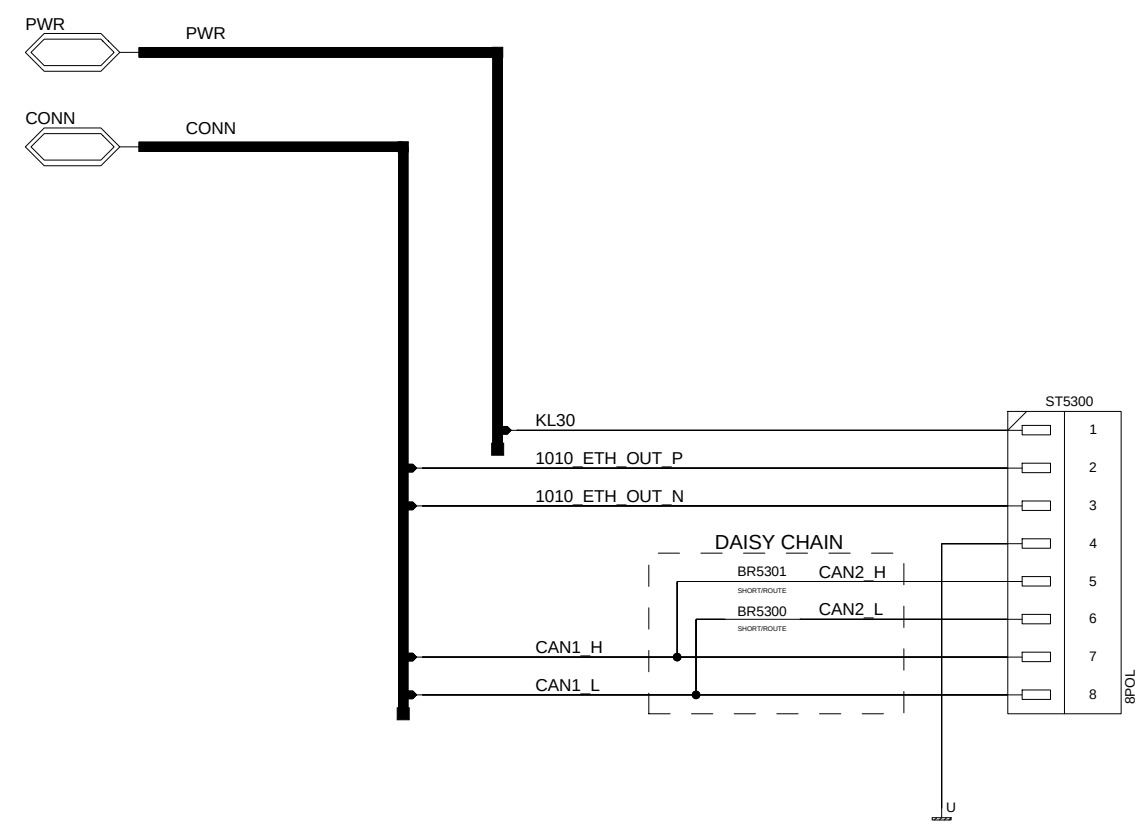
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			Copyright reserved	Wiring modifications reserved		date		name		
					created					
					checked					
schematic						Sheet	Sheets of		Sample	
FPGA_POWER						9	31			
substitute for:						Type	Variant		Part Number	
						C143				
substitute by:										



RH550 INTERRUPT TABLE		
INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	
INTP2	ISO	
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	

Rev.	type of change		change no.	date of change	auth. checked
		Copyright reserved	date		name
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			checked		
schematic: 4010_IDC			Sheet 11	Sheets of 31	Sample
			Type Variant C143	Part Number	
substitute for:		substitute by:			

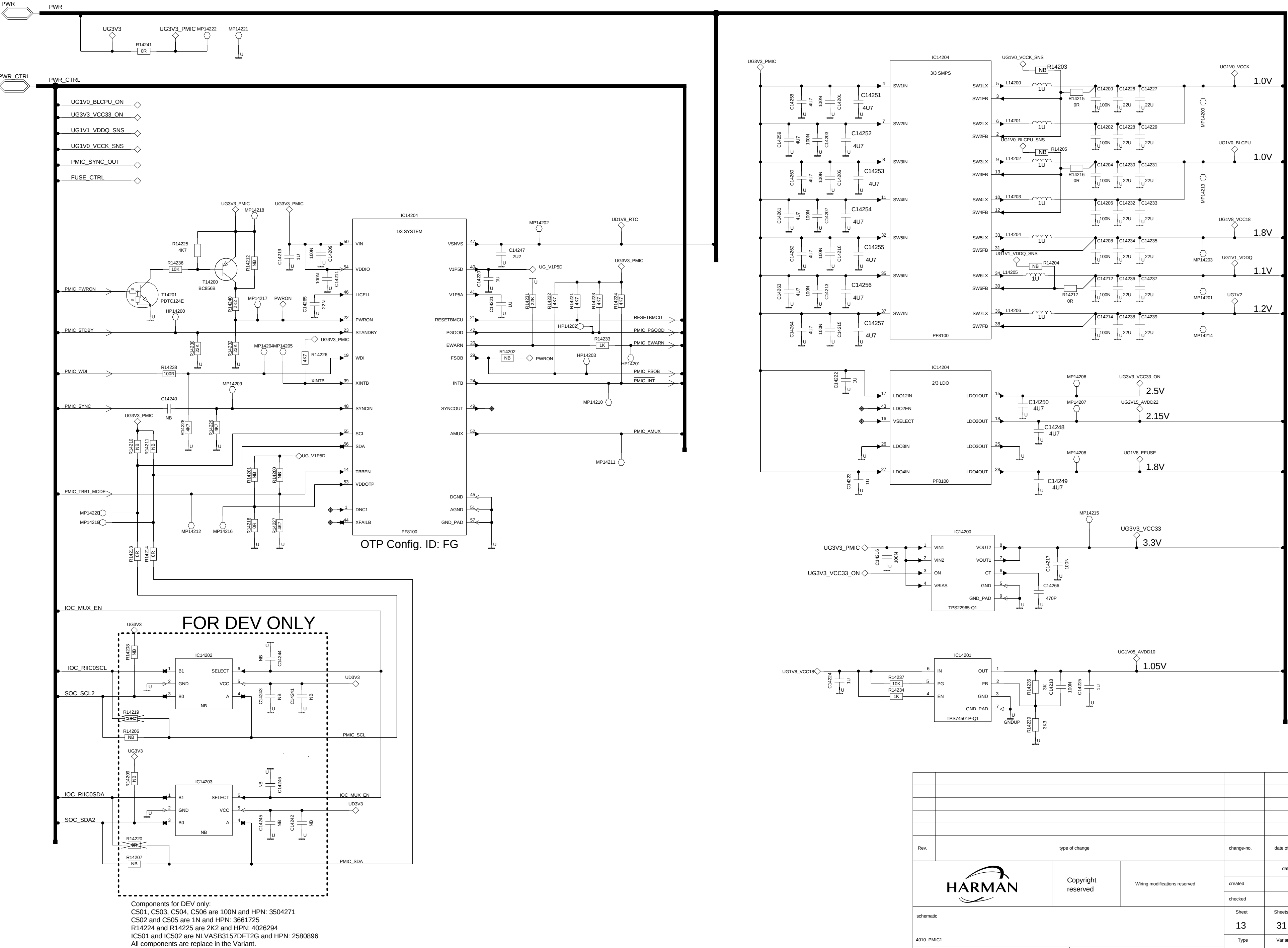
MAIN CONNECTOR (5300-5399)



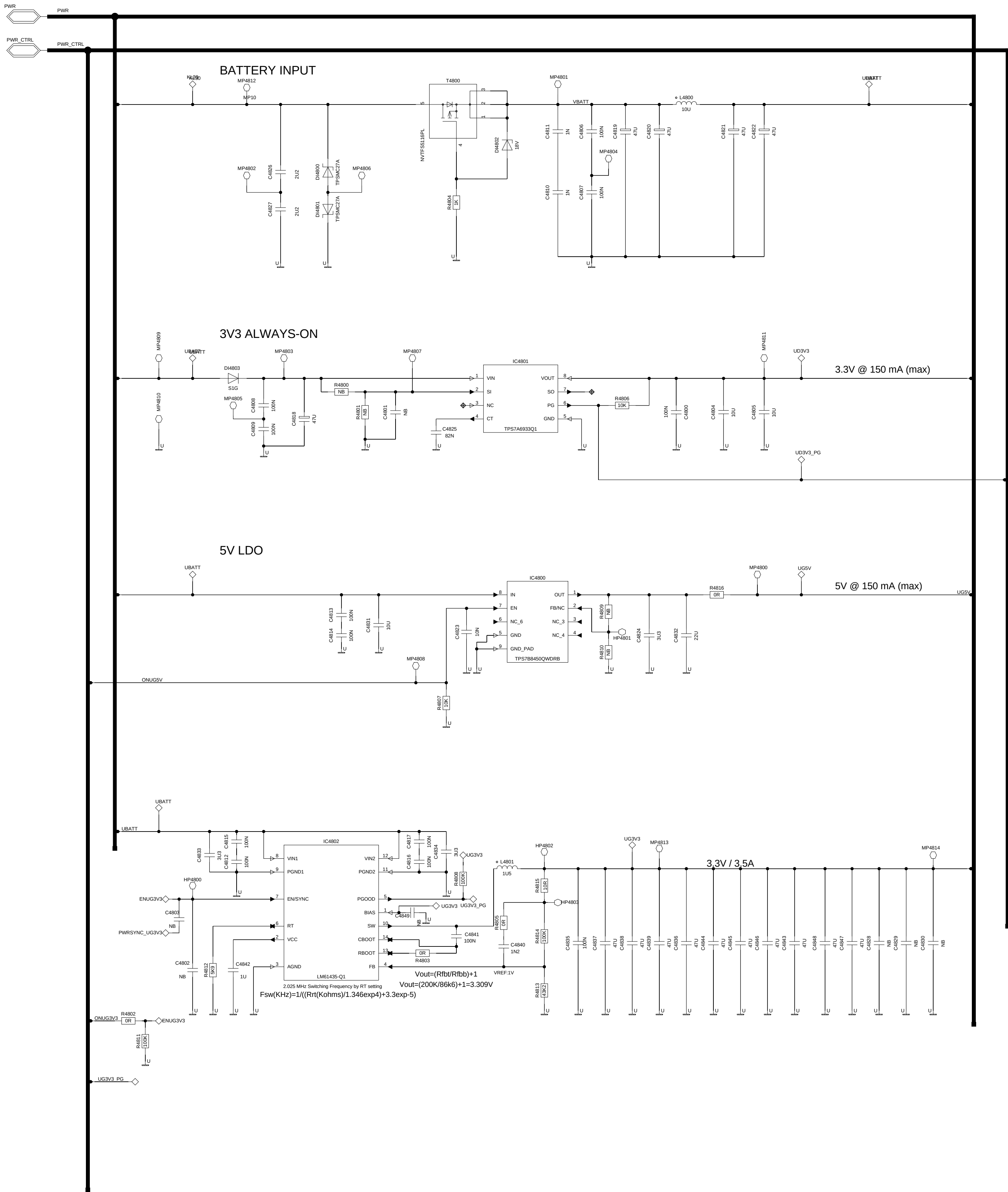
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				created		
				checked		
schematic			Sheet	Sheets of		Sample
4010_Main_Connector			12	31		
substitute for:			Type	Variant		Part Number
substitute by:						

PMIC1 (14200-14299)

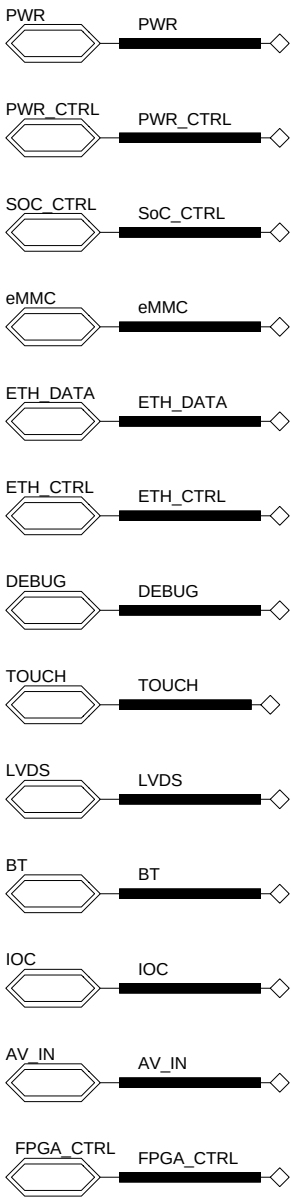


Rev.	type of change		change-no.	date of change	auth. checked
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			created		
			checked		
schematic			Sheet	Sheets of	Sample
4010_PMIC1			13	31	
substitute for:			Type	Variant	Part Number
substitute by:					

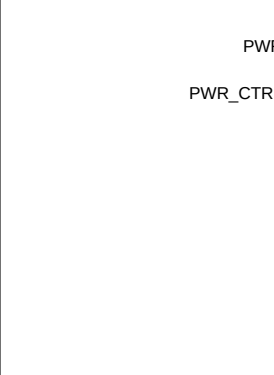


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				created					
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schematic				Sheet 14	Sheets of 31		Sample		
4030_Power				Type	Variant		Part Number		
substitute for:			substitute by:						

SOC (10000-10999)

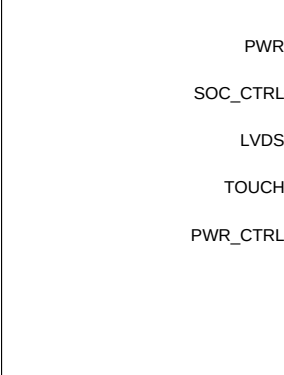


4010 SoC Power



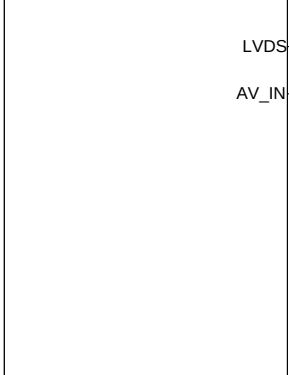
10500 - 10599

4010 SoC_CTRL



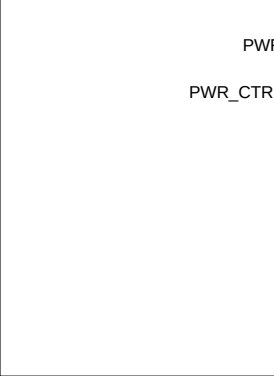
10600 - 10699

4010 SoC Video



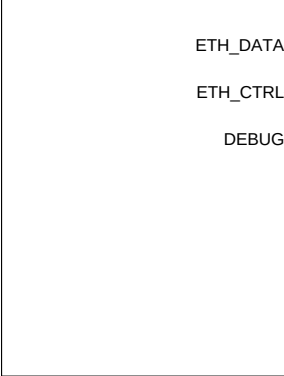
10700 - 10799

4010 SoC LPDDR4



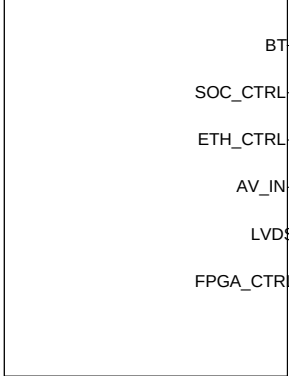
10400 - 10499

4010 SoC_ETH_PCIE_USB



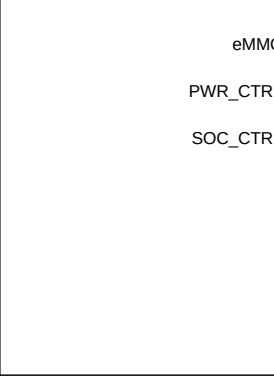
10300 - 10399

4010 SoC Audio



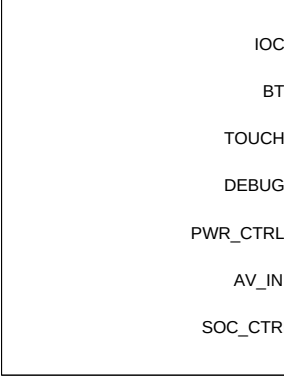
10800 - 10899

4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

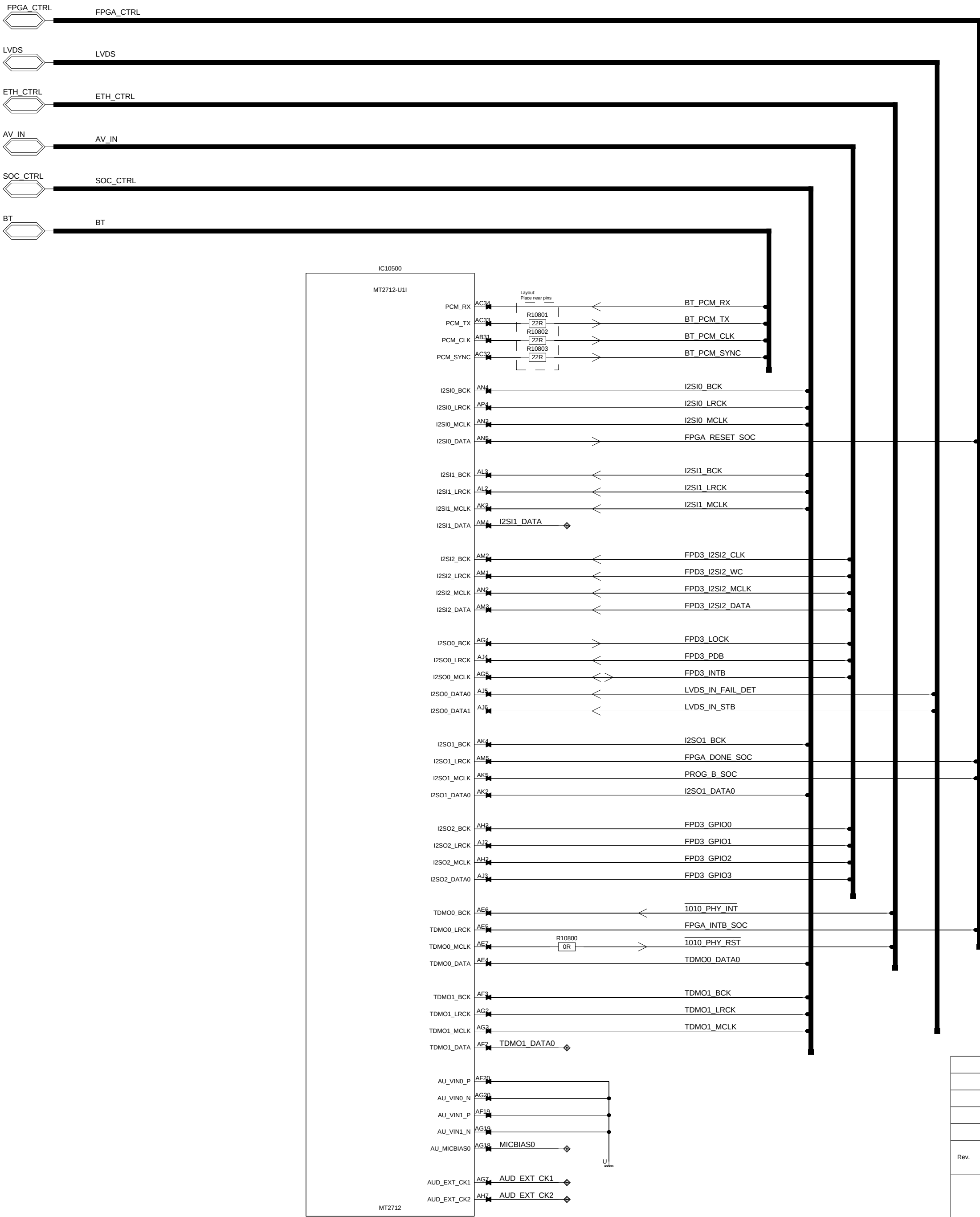


10200 - 10299

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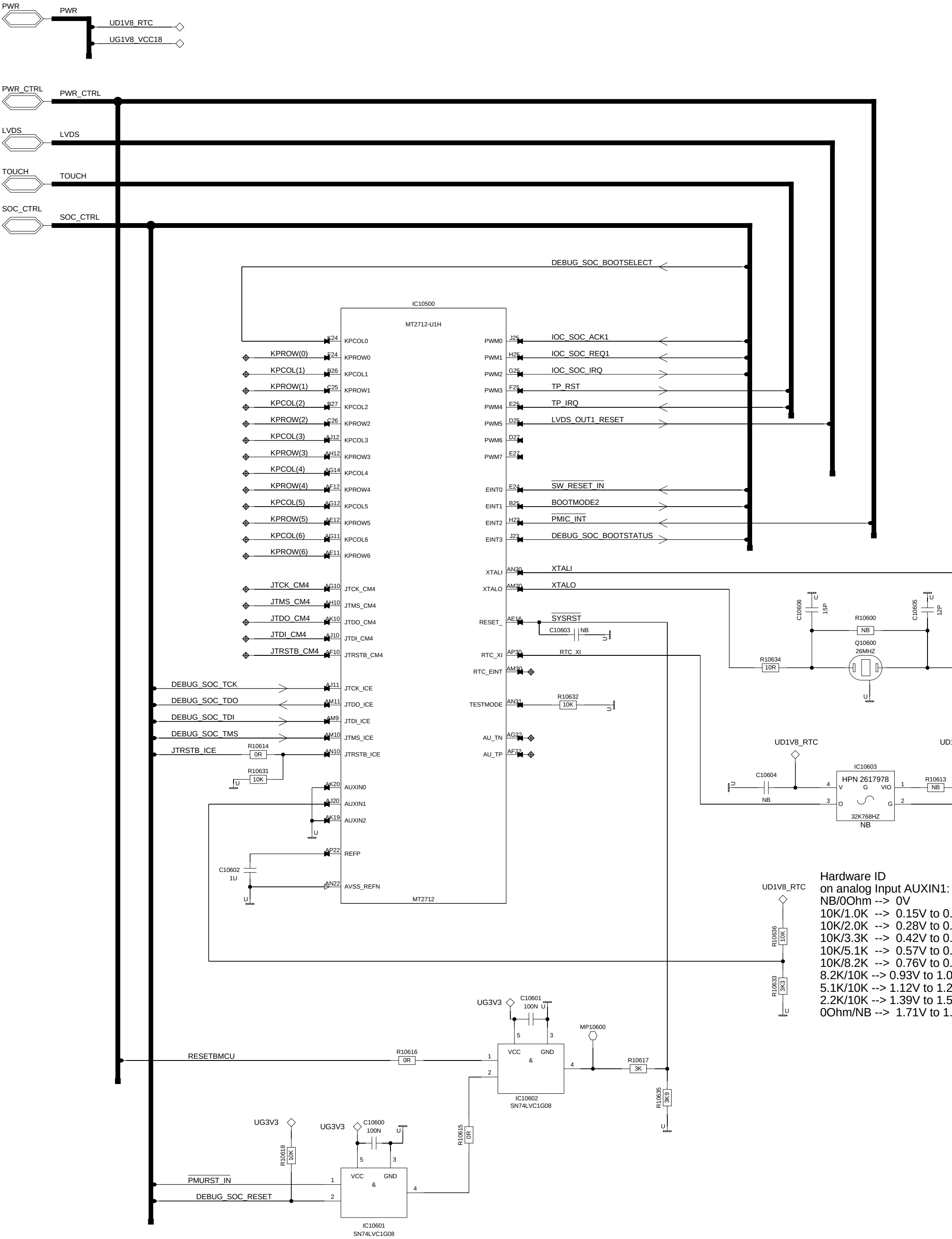
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		Copyright reserved	Wiring modifications reserved	date		name
				created		
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schematic			Sheet	Sheets of		Sample
4010_SoC			15	31		
substitute for:			Type	Variant		Part Number
substitute by:						

SOC_AUDIO (10800-10899)



Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Writing modifications reserved		date	name
		created			
		checked			
schematic			Sheet	Sheets of	
4010_SoC_Audio			16	31	
substitute for:				Type	Variant
				Part Number	

SOC_CTRL (10600-10699)



SOC_CTRL						
Pin Strapping						
Mode	I2SI0_LRCK	I2SI0_MCLK	I2SI0_BCK	I2SI1_BCK	I2SI1_LRCK	I2SI1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIe	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIe	X	X	X	X	0	X
PCIIE Port0 for RC	X	X	X	1	X	X
PCIIE Port0 for EP	X	X	X	0	X	X
PCIIE Port1 for RC	X	X	1	X	X	X
PCIIE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

IPC1POR/Boot Mode					
I2SO1_DATA0 status[1]	L	L	H	H	
I2SO1_BCK status[0]	L	H	L	H	
1	EMMC	NAND	NOR	SD	
2	SD	EMMC	NAND	NOR	
3	NOR	SD	EMMC	NAND	
4	NAND	NOR	SD	EMMC	

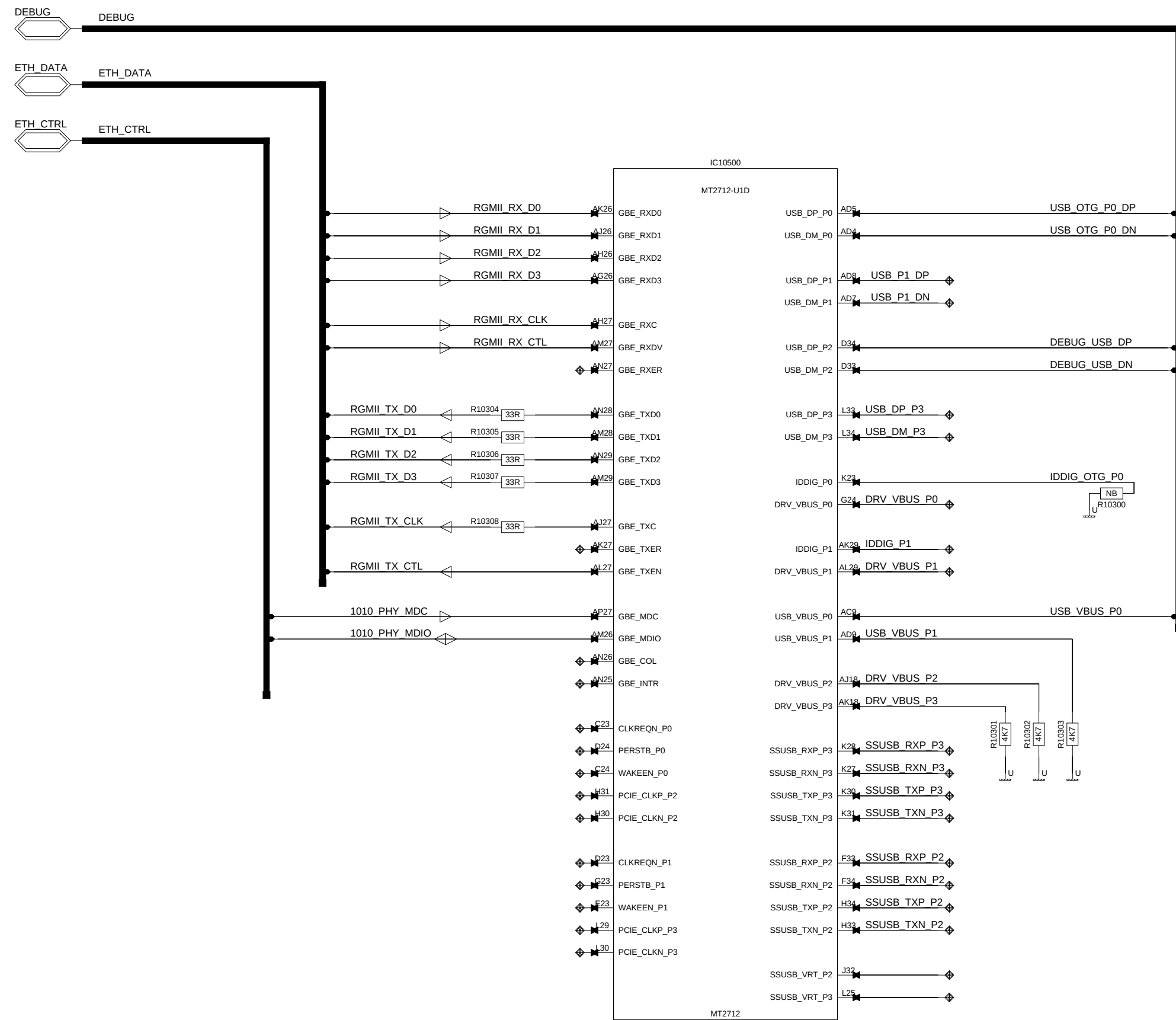
Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO0_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

Hardware ID
on analog Input AUXIN1:
NB/OOhm --> 0V
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
0Ohm/NB --> 1.71V to 1.89V

ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

Rev.	type of change		change-no.	date of change	auth.	checked
 Copyright reserved Wiring modifications reserved				date	name	
			created			
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schematic			Sheet	Sheets of	Sample	
4010_Soc_CTRL			17	31		
substitute for:			Type	Variant	Part Number	
substitute by:						

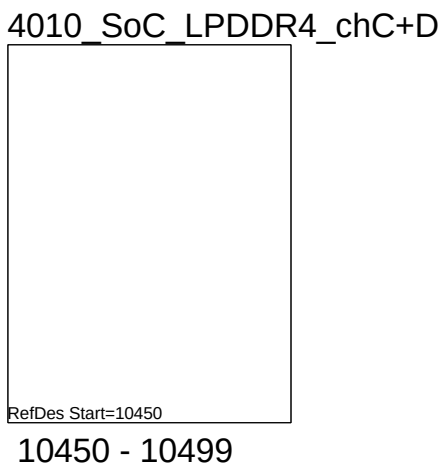
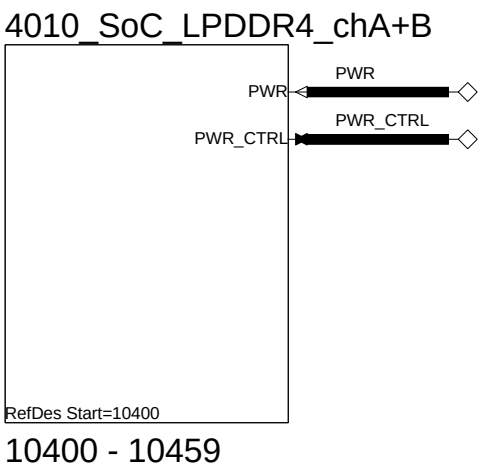
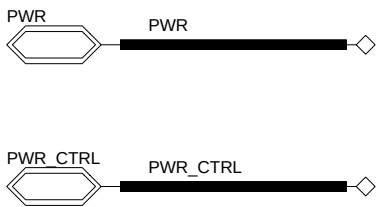
SOC_ETH_PCl_e_USB (10300-10399)



Rev.	type of change					change-no.	date of change	auth.	checked
			Copyright reserved	Writing modifications reserved	date		name		
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schematic 4010_Soc_ETH_PCIE_USB						Sheet 18	Sheets of 31	Sample	
						Type	Variant	Part Number	
substitute for:			substitute by:						

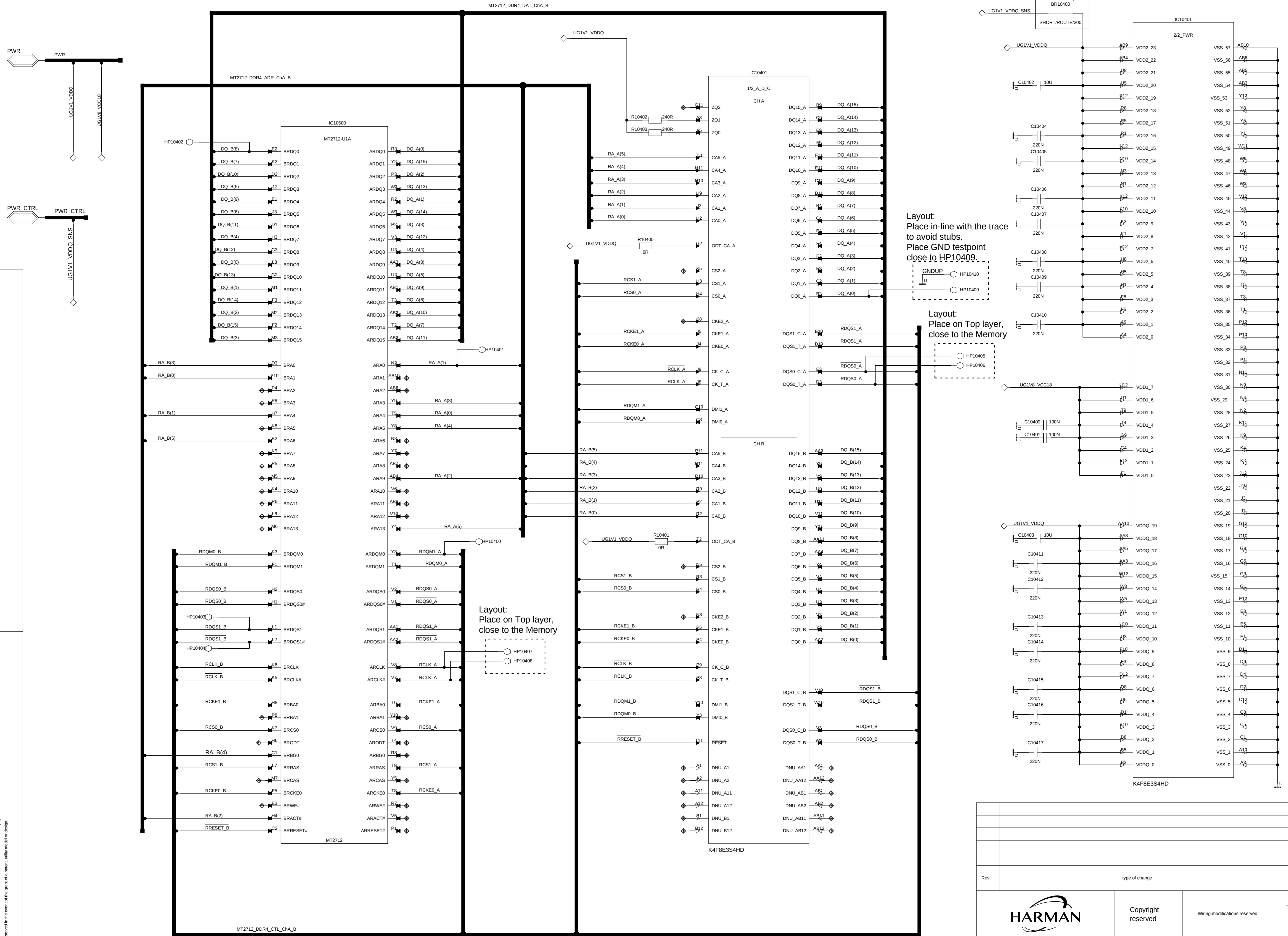
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SOC_LPDDR4 (10400-10499)



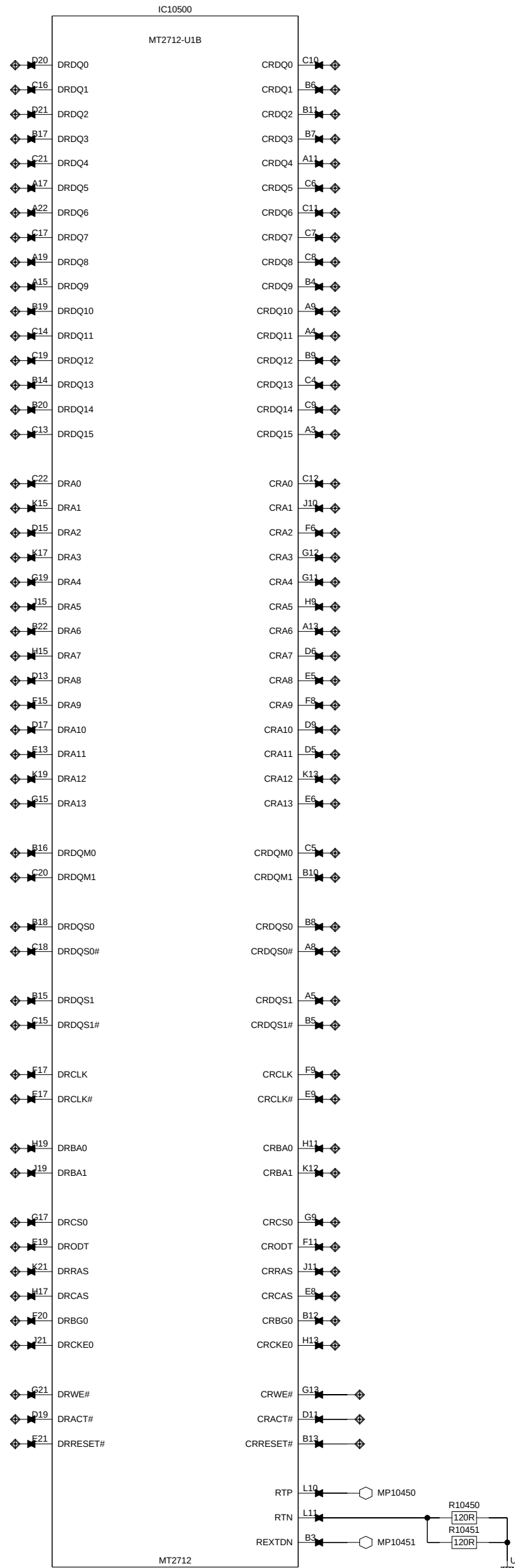
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				created		
				checked		
schematic			Sheet	Sheets of		Sample
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant		Part Number

SOC_LPDDR4 (10400-10449)



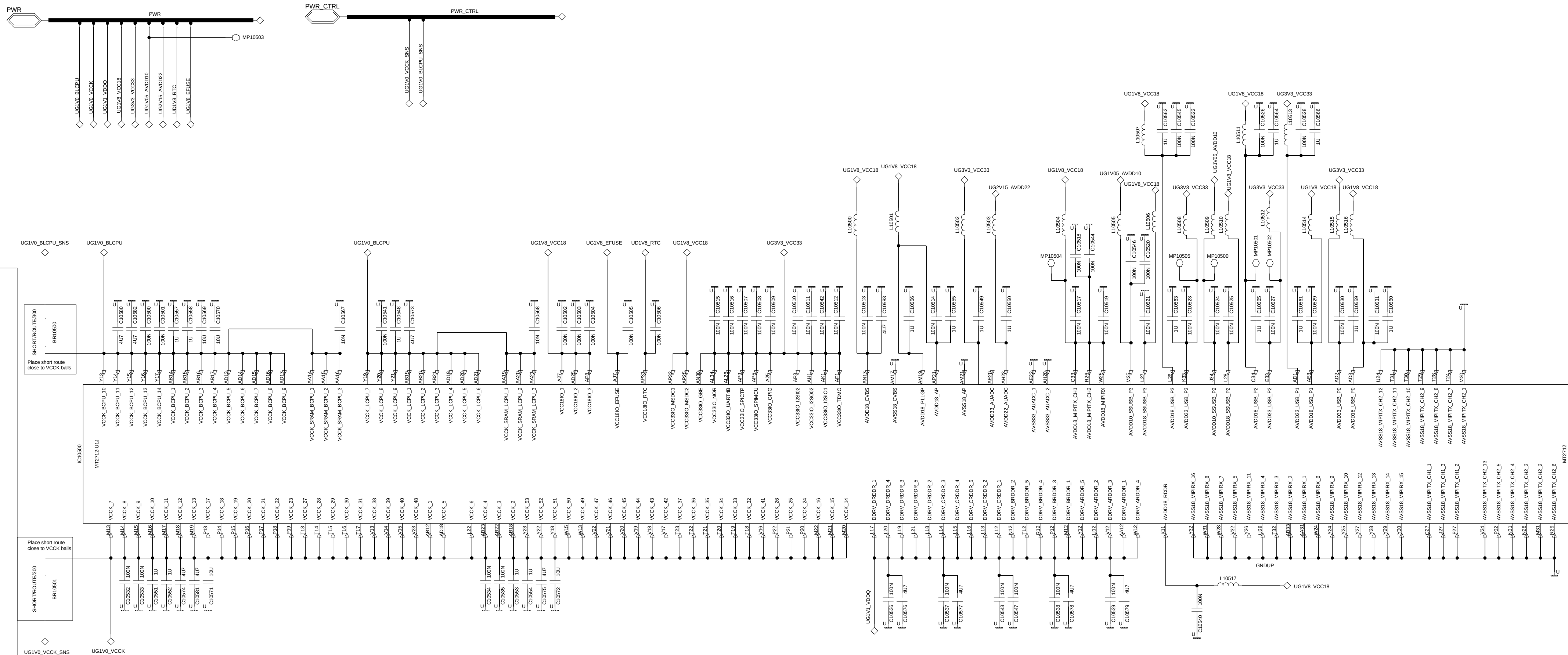
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SOC_LPDDR4 (10450-10499)

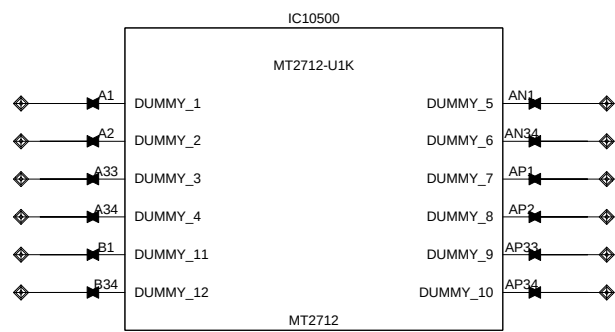


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				checked		
schematic				Sheet	Sheets of	Sample
4010_SoC_LPDDR4_chC+D				21	31	
substitute for:		substitute by:		Type	Variant	Part Number

SOC_POWER (10500-10599)



4010 SoC GND



Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Wiring modifications reserved	date		name
			created		
			checked		
schematic		Sheet 23	Sheets of 31	Sample	
4010_SoC_GND		Type	Variant	Part Number	
substitute for:		substitute by:			

The diagram illustrates the internal architecture of the MT2712-U1E SoC and its connections to various external components. The SoC is shown as a large rectangle with pins on its left and right sides. Internal components include multiple I2C controllers (I2C_SDA0 to I2C_SCL5), UARTs (URXD0 to URXD5, UTXD0 to UTXD5), SPI controllers (SPI0_MI to SPI5_CSN), and various peripheral controllers (SOCFPGA, IOC, BT, DEBUG, AV_IN, PWR_CTRL, TOUCH). External components include a TP (TP_SDA, TP_SCL), FPD3 (FPD3_I2C5_SDA, FPD3_I2C5_SCL), and various resistors (R10200, R10201, R10202). The diagram shows the connection of these components to the SoC pins and to each other.

Internal Components and Connections:

- I2C Controllers:** I2C_SDA0 to I2C_SCL5. Connections include TP_SDA, TP_SCL, SOC_SDA2, SOC_SCL2, FPD3_I2C5_SDA, and FPD3_I2C5_SCL.
- UARTs:** URXD0 to URXD5, UTXD0 to UTXD5. Connections include DEBUG_SOC_UART_RXD, DEBUG_SOC_UART_TXD, BT_URXD3, BT_UTXD3, BT_URTS3, and BT_UCTS3.
- SPI Controllers:** SPI0_MI to SPI5_CSN. Connections include SOCFPGA_MISO_EEPROM, SOCFPGA_MOSI_EEPROM, SOCFPGA_SCK_EEPROM, and SOCFPGA_CSB_EEPROM.
- Other Components:** SOC_CTRL, IOC, BT, DEBUG, AV_IN, PWR_CTRL, TOUCH, TP, FPD3, and various resistors (R10200, R10201, R10202).

External Connections:

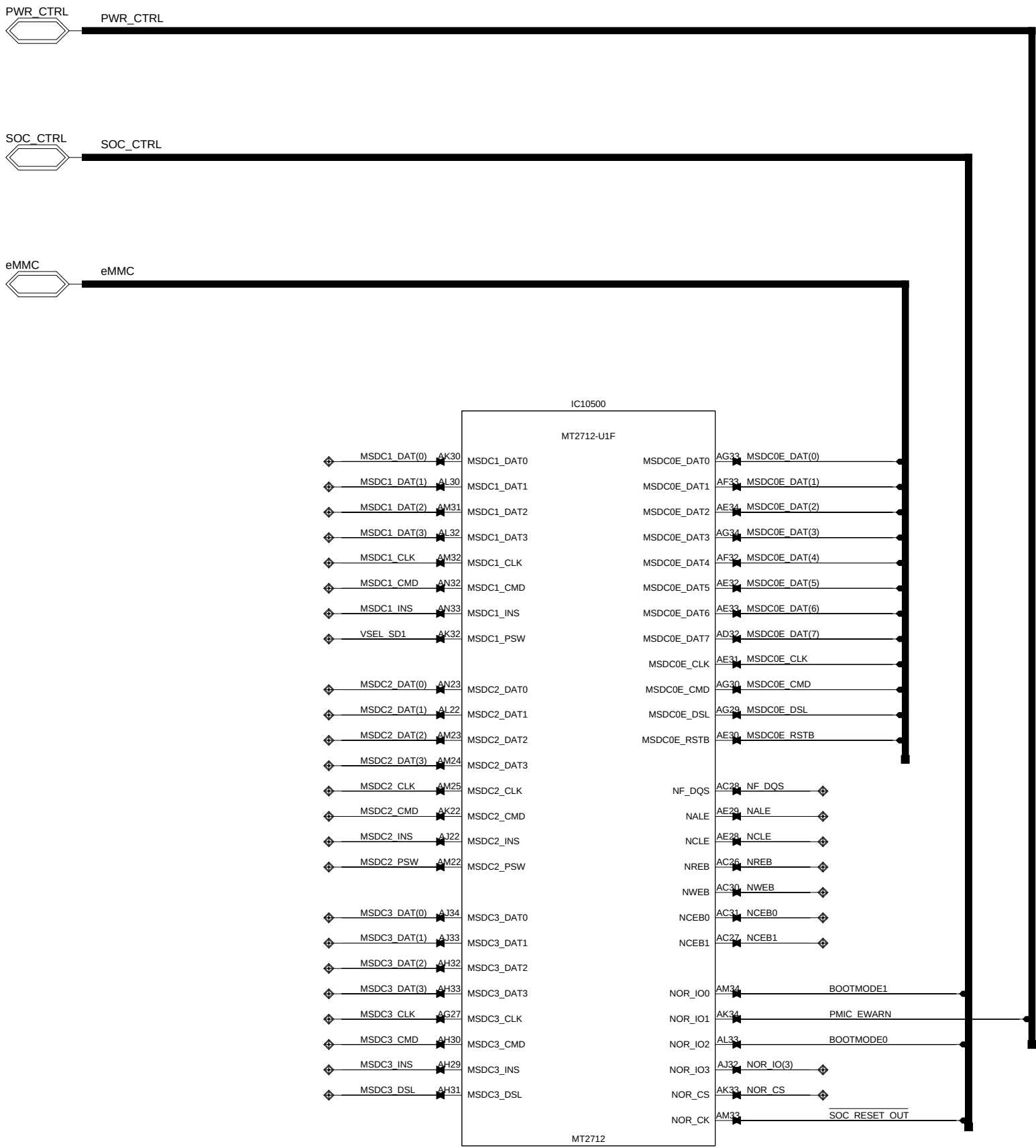
- SOC_CTRL:** Connected to the top of the SoC.
- IOC:** Connected to the top of the SoC.
- BT:** Connected to the top of the SoC.
- DEBUG:** Connected to the top of the SoC.
- AV_IN:** Connected to the top of the SoC.
- PWR_CTRL:** Connected to the top of the SoC.
- TOUCH:** Connected to the top of the SoC.
- TP:** Connected to the top of the SoC.
- FPD3:** Connected to the top of the SoC.
- BT_URXD3, BT_UTXD3, BT_URTS3, BT_UCTS3:** Connected to the top of the SoC.
- SOCFPGA_MISO_EEPROM, SOCFPGA_MOSI_EEPROM, SOCFPGA_SCK_EEPROM, SOCFPGA_CSB_EEPROM:** Connected to the top of the SoC.
- IOC_SPI4_MI, IOC_SPI4_MO, IOC_SPI4_CK, IOC_SPI4_CS:** Connected to the top of the SoC.

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 24	Sheets of 31	Sample
4010_SoC_Serial_IF			Type	Variant	Part Number
substitute for:		substitute by:			

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

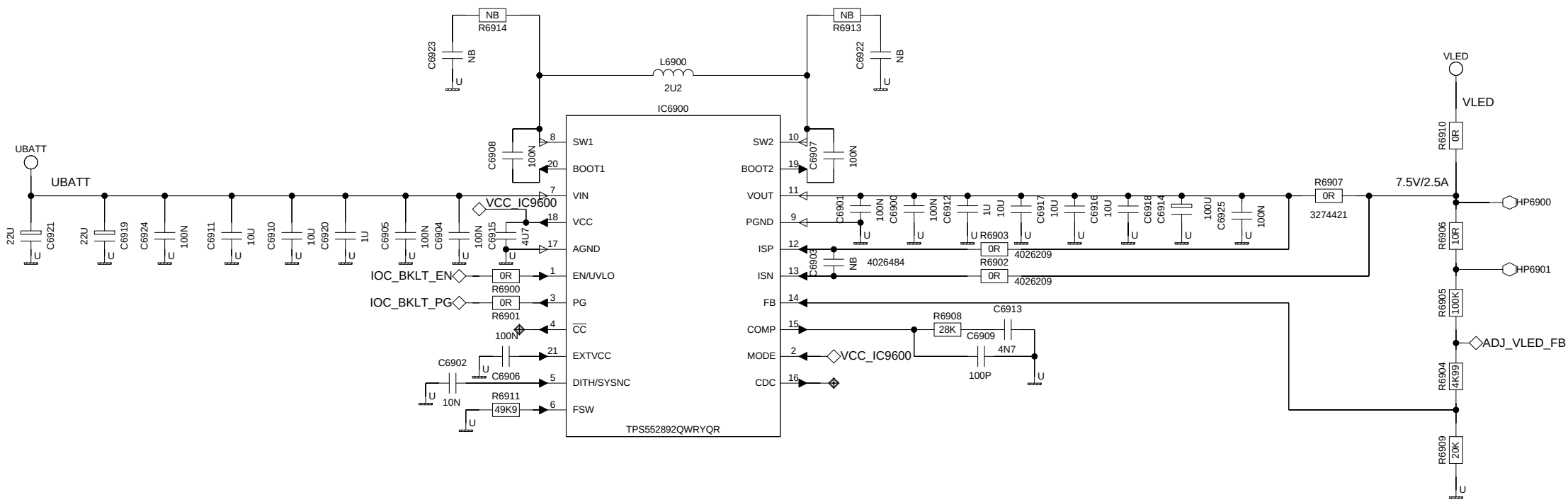
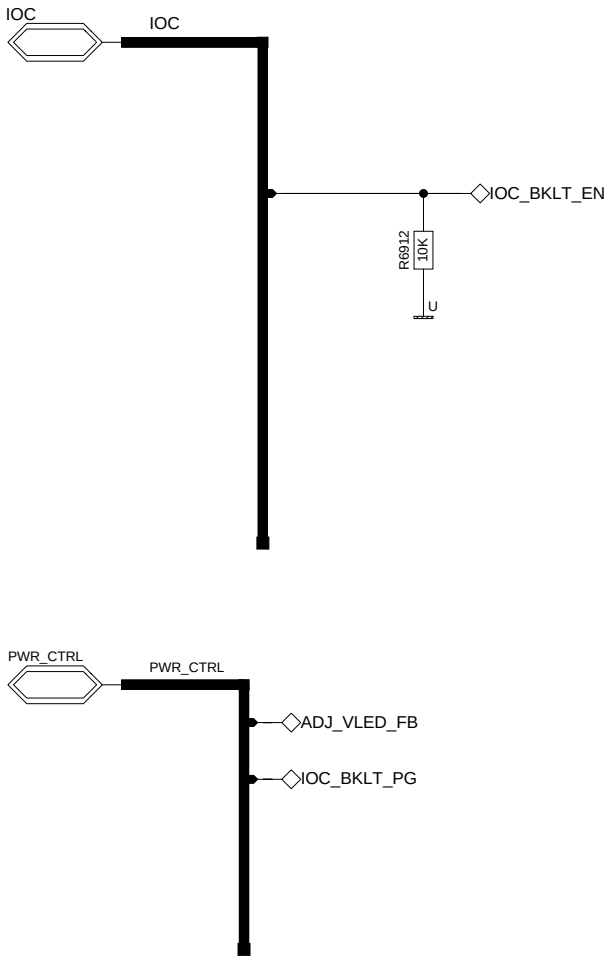
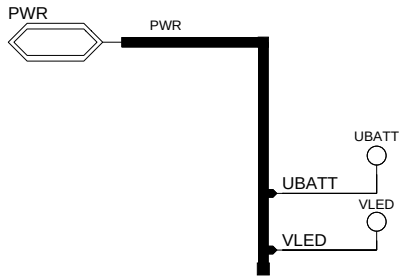
Rev.	type of change					change-no.	date of change		auth.	checked
			Copyright reserved	Writing modifications reserved	date		name			
					created					
					checked					
schematic						Sheet	Sheets of		Sample	
4010_Soc_Video						25	31			
substitute for:				substitute by:		Type	Variant		Part Number	

SOC_eMMC_NAND_SID (10100-10199)



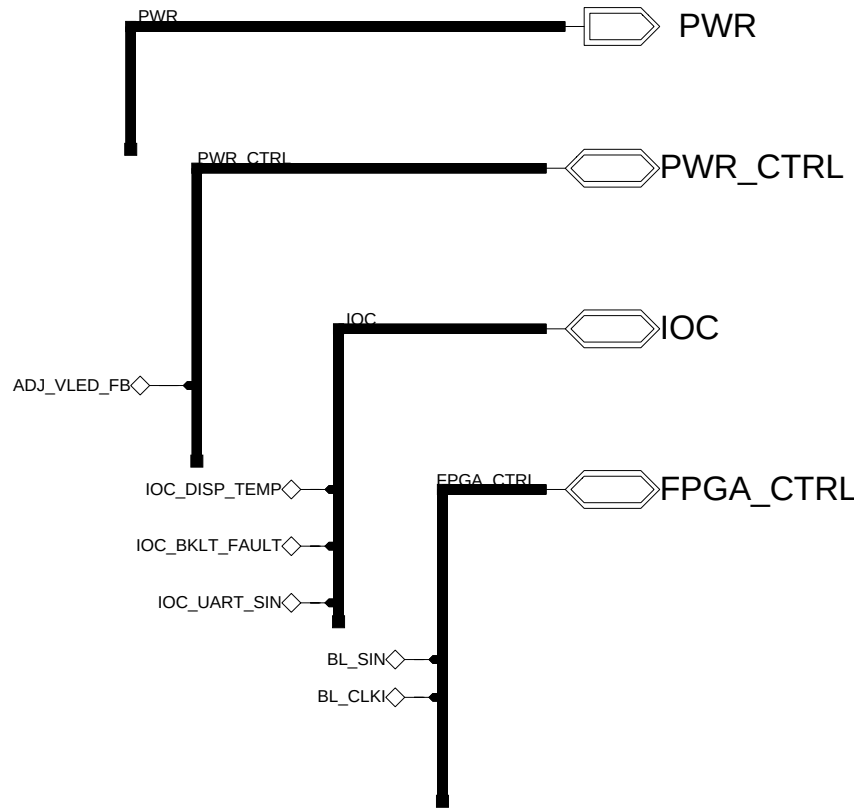
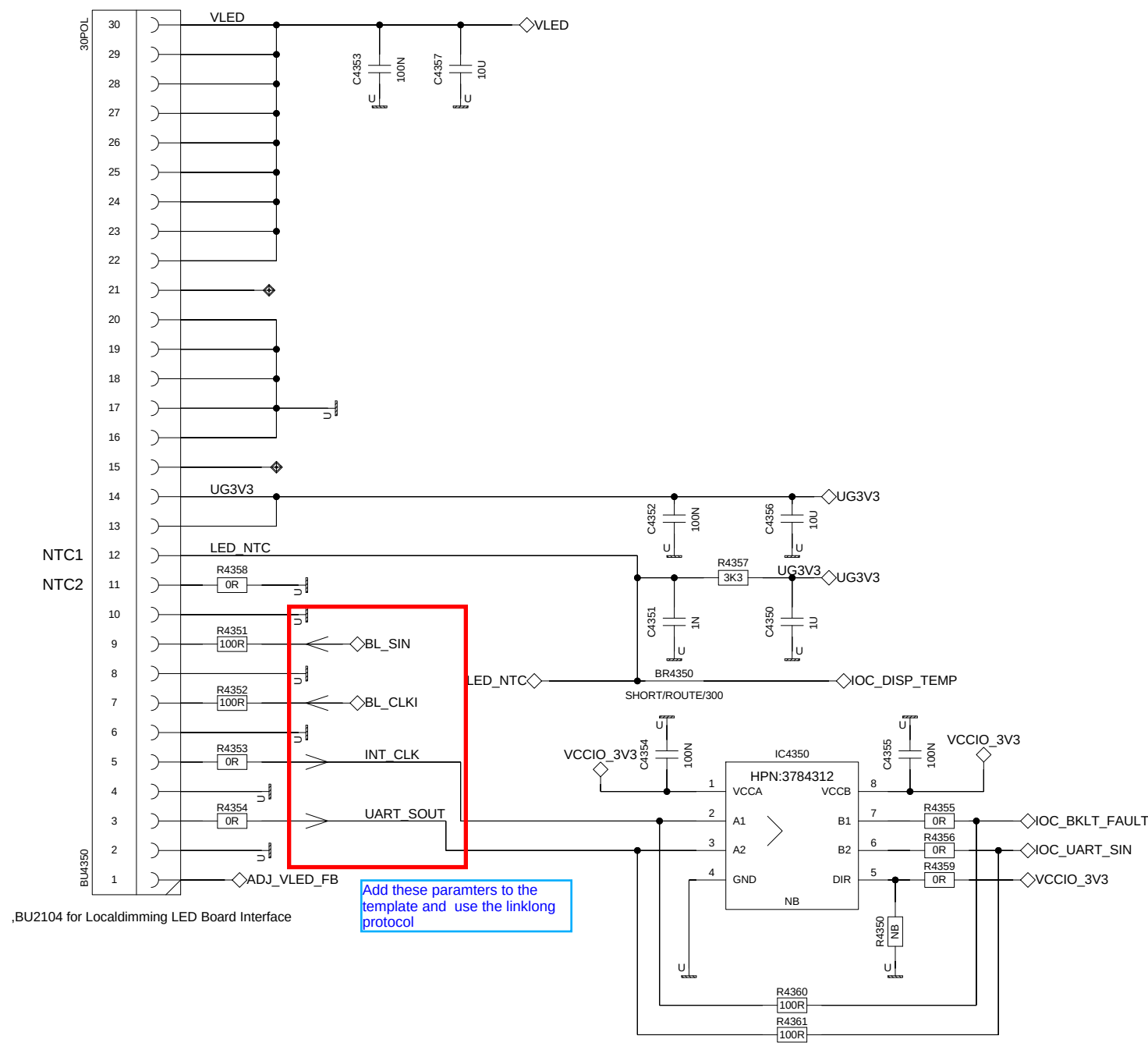
Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet 26	Sheets of 31	Sample
4010_SoC_eMMC_NAND_SD				Type	Variant	Part Number
substitute for:		substitute by:				

TFT BL (6900-6999)

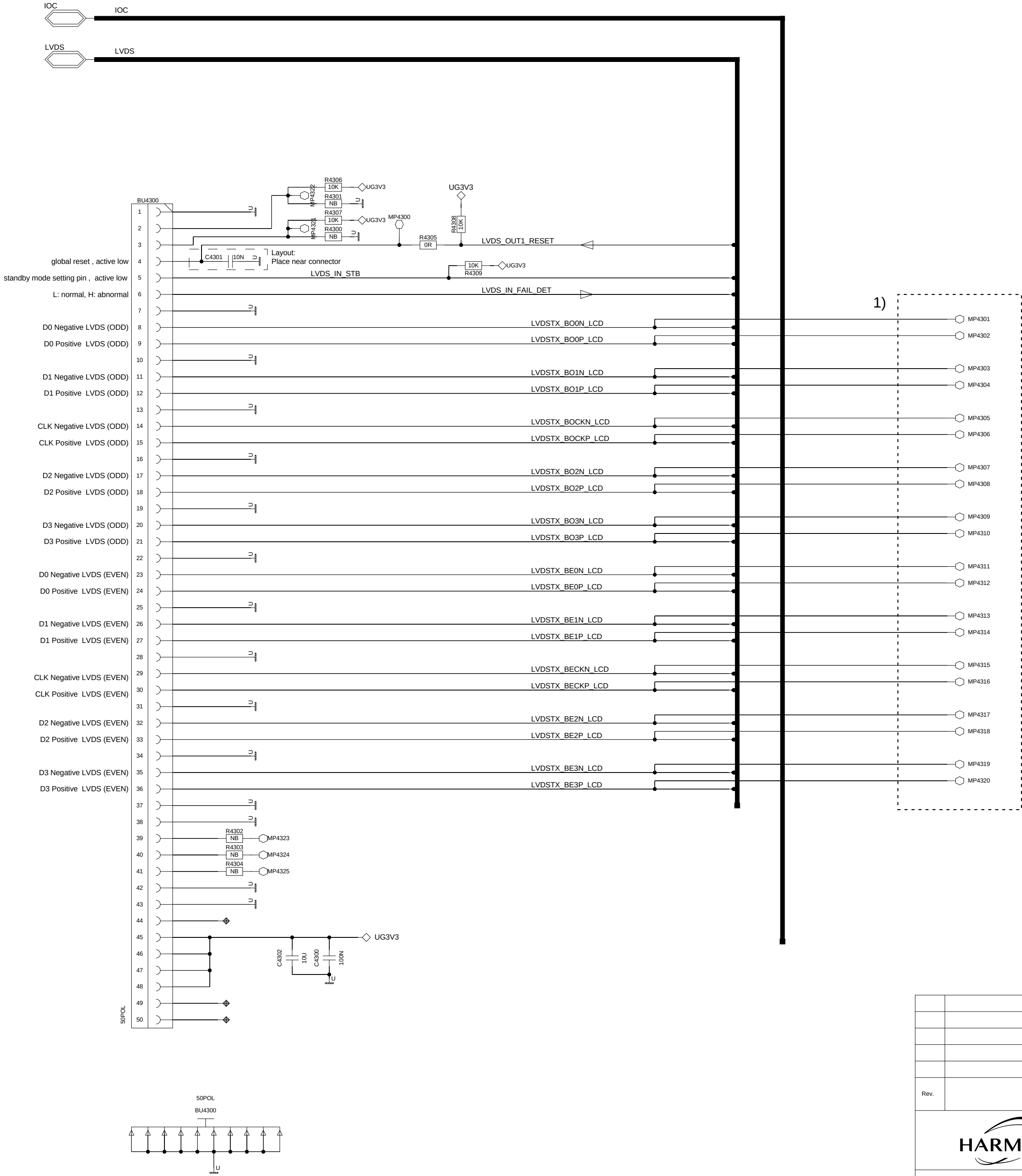
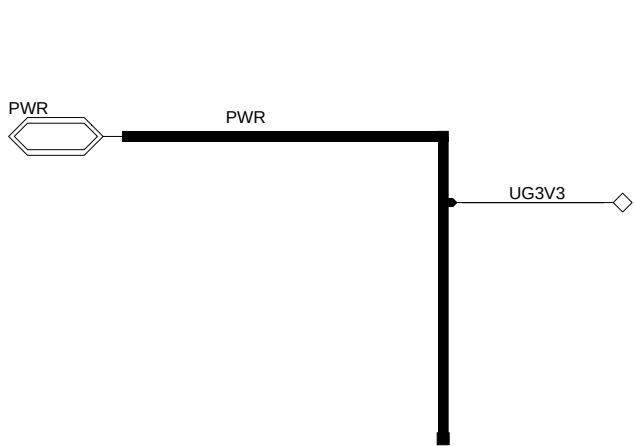


Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Wiring modifications reserved	date		name
			created		
			checked		
schematic		Sheet 27	Sheets of 31	Sample	
4010_TFT_BL		Type	Variant	Part Number	
substitute for:	substitute by:		C143		

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	Copyright reserved	date		name	
		created			
		checked			
schematic	Sheet		Sheets of		Sample
4010_TFT_Bl_Conn	28		31		
substitute for:	Type		Variant		Part Number
	substitute by:		C143		

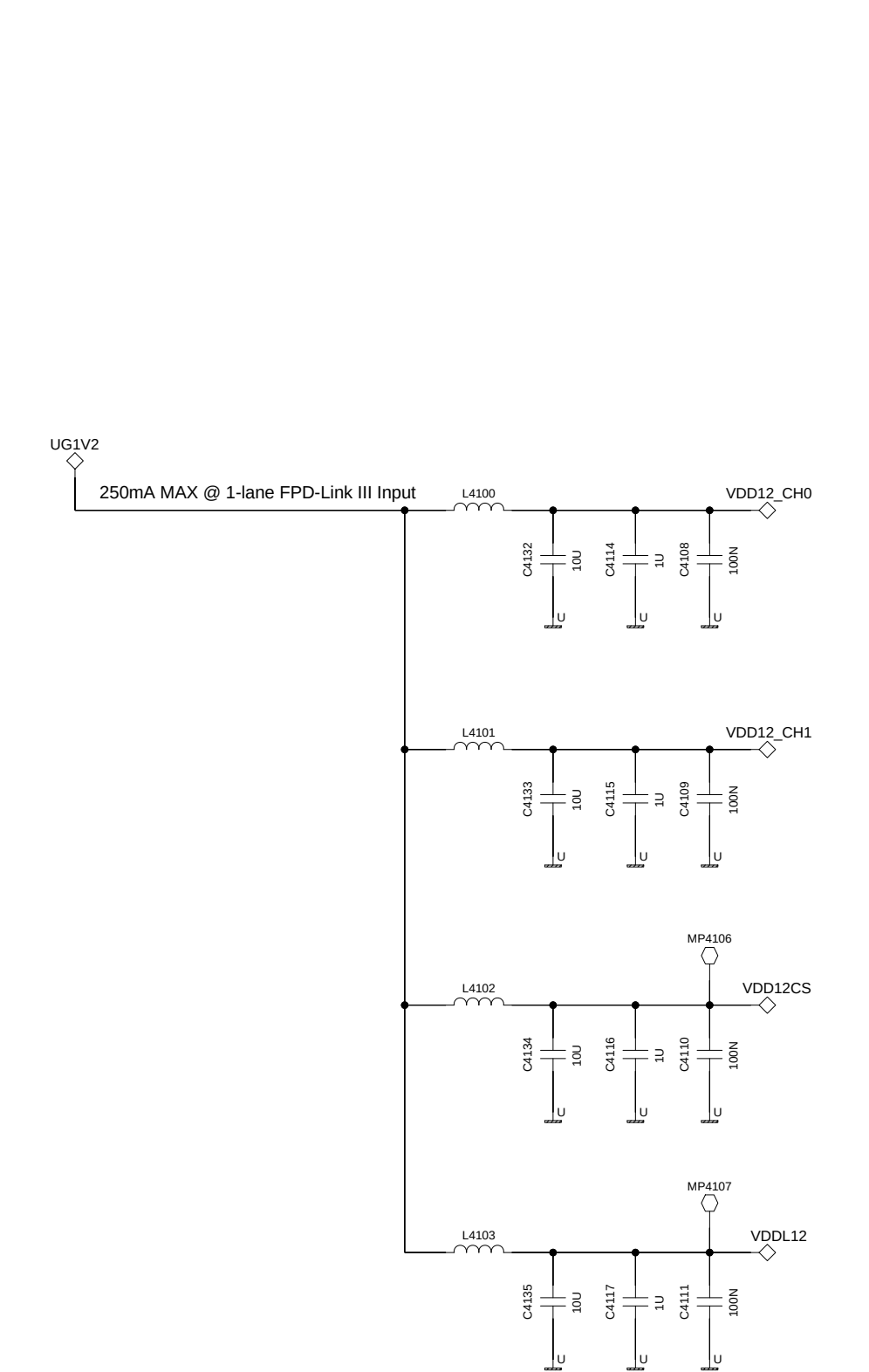


Verify pin out during layout

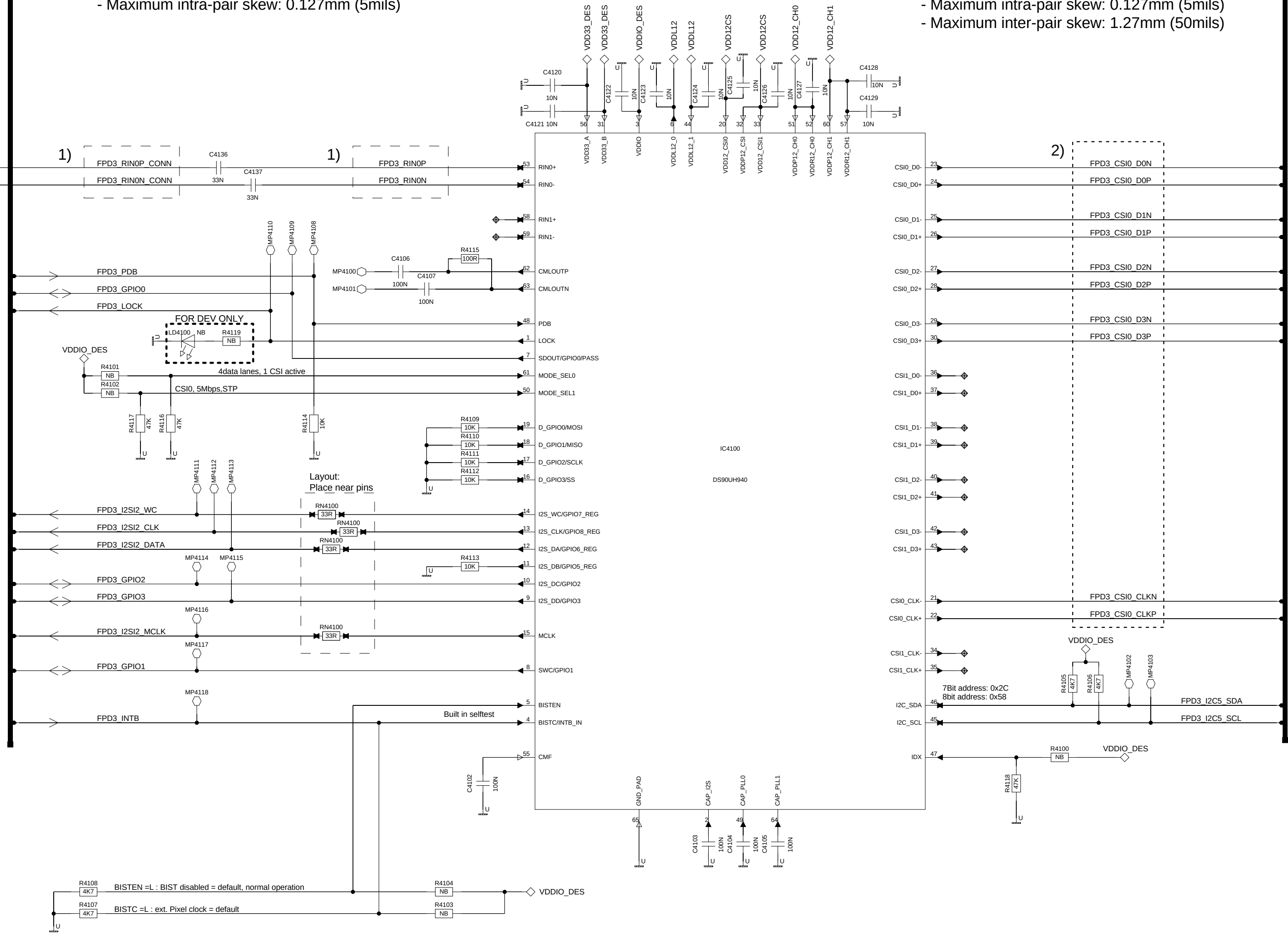
1) Testpoints should be placed as close as possible to an existing VIA

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 29	Sheets of 31	Sample
4010_TFT_Conn			Type	Variant C143	Part Number
substitute for:		substitute by:			

The schematic diagram illustrates the input sections of the MP4105. The AV_IN section features a 6mA MAX @ 1-lane FPD-Link III Input connected to VDDIO_DES, with bypass capacitors C4130, C4112, C4100, C4118, and C4105. The PWR section features a 122mA MAX @ 1-lane FPD-Link III Input connected to VDD33_DES, with bypass capacitors C4131, C4113, C4101, and C4119. A BU4100 component is also shown.

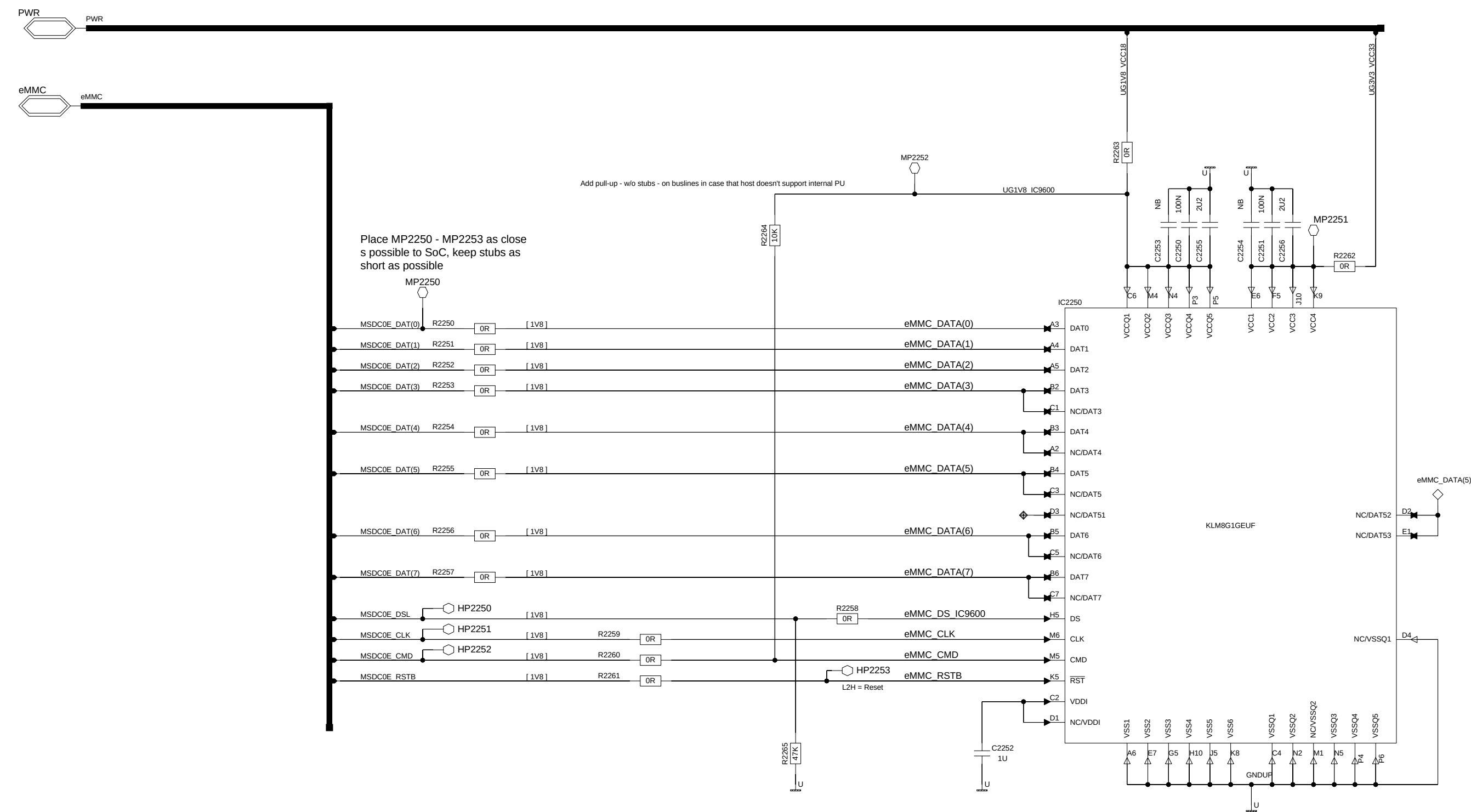


- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum inter-pair skew: 1.27mm (50mils)



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		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 30	Sheets of 31	Sample
4010_VIDEO_DES			Type	Variant	Part Number
substitute for:		substitute by:		C143	

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617